



User Manual **Model 765**

Fast Rise Time Pulse Generator



Summary

Model 765 REV.B USER Manual Cover.....	1
GENERAL SAFETY SUMMARY	4
TO AVOID FIRE OR PERSONAL INJURY	4
<i>Use Proper Power Cord.....</i>	4
<i>Ground the Product</i>	4
<i>Observe All Terminal Ratings.....</i>	4
<i>Power Disconnect</i>	4
<i>Do Not Operate Without Covers.....</i>	4
<i>Do Not Operate With Suspected Failures</i>	4
<i>Avoid Exposed Circuitry</i>	4
<i>Do Not Operate in Wet/Damp Conditions.....</i>	4
<i>Do Not Operate in an Explosive Atmosphere.</i>	4
<i>Keep Product Surfaces Clean and Dry.....</i>	4
<i>Provide Proper Ventilation</i>	5
SAFETY SYMBOLS	6
OPERATING ENVIRONMENT	7
AC POWER SOURCE	8
POWER CONSUMPTION.....	8
CALIBRATION	8
CLEANING	8
ABNORMAL CONDITIONS.....	8
ENVIRONMENTAL CONSIDERATIONS.....	9
PRODUCT END-OF-LIFE HANDLING	9
EQUIPMENT RECYCLING	9
OPERATING REQUIREMENTS	10
POWER SUPPLY	10
MECHANICAL CHARACTERISTICS	10
ENVIRONMENTAL CHARACTERISTICS	10
PREFACE	11
PACKAGE CONTENTS.....	11
2-4 CHANNELS.....	12
TRIGGER, VIEW, GENERATE AND SYNC	12
TOUCH SCREEN DISPLAY AND SOFT KEYBOARD	12
SIMPLERIDER PULSE TOUCH USER INTERFACE	12
RECOMMENDED ACCESSORIES.....	13
POWER THE INSTRUMENT ON AND OFF	13
PROTECT YOUR INSTRUMENT FROM MISUSE	13
OBTAINING THE LATEST VERSION RELEASES	15
INSTALL PULSE GENERATOR Model 765 APPLICATION	16
REMOTE CONTROL	18
PREREQUISITE	18
AT INSTRUMENT COMMUNICATOR.....	23
INSTRUMENT OVERVIEW.....	25

FRONT PANEL	25
REAR PANEL	26
GETTING STARTED WITH Model 765.....	27
INSTRUMENT CONTROL	27
<i>Front Panel Buttons</i>	28
<i>Front Panel Numeric Keypad</i>	30
OUTPUT AND PULSE DEFINITION	32
OUTPUT CHANNEL PARAMETERS	33
<i>Vertical Parameters</i>	33
<i>Period / Frequency</i>	34
PULSE PARAMETERS.....	35
<i>Width / Duty Cycle</i>	35
<i>Delay / Phase</i>	35
PARAMETER CONFLICTS	37
PARAMETER LIMITS	37
MULTIPLE PULSES OPERATION	37
<i>Hint: how to generate a clock</i>	40
TRIGGER CONTROLS	42
TRIGGER OUTPUT	42
TRIGGER INPUT	42
<i>Trigger Modes</i>	43
<i>Multiple Pulse Interactions</i>	51
QUICK START	53
SINGLE PULSE	53
DOUBLE PULSE	54
Model 765 SOFTWARE.....	57
USER INTERFACE DESCRIPTION	57
<i>Parameters Area</i>	59
<i>Graph Area</i>	61
<i>Command Bar Area</i>	62
CONFIGURATIONS	65
SAVE AS...	65
EXPORT CONFIGURATION.....	65
LOAD FROM...	66
SETTINGS	67
TRIGGER SETUP.....	67
OUTPUT SETTINGS.....	69
CLOCK SETTINGS	73
REMOTE / LOCALE MODE	74
<i>Remote Desktop Connection</i>	75
<i>Self Calibration and Diagnostic</i>	75
<i>Self Calibration</i>	76
<i>Self Diagnostic</i>	76
<i>Quick tips</i>	77

General Safety Summary

Review the following safety precautions to avoid injury and prevent damage to this product or any products connected to it.

To avoid potential hazards, use this product only as specified.

Only qualified personnel should perform service procedures.

To Avoid Fire or Personal Injury

Use Proper Power Cord

Use only the power cord specified for this product and certified for the country of use.

Ground the Product

This product is grounded through the grounding conductor of the power cord. To avoid electric shock, the grounding conductor must be connected to earth ground. Before making connections to the input or output terminals of the product, ensure that the product is properly grounded.

Observe All Terminal Ratings

To avoid fire or shock hazard, observe all ratings and markings on the product. Consult the product manual for further ratings information before making connections to the product.

Power Disconnect

The power cord provides Mains disconnect.

Do Not Operate Without Covers

Do not operate this product with covers or panels removed.

Do Not Operate With Suspected Failures

If you suspect that there is damage to this product, have it inspected by qualified service personnel.

Avoid Exposed Circuitry

Do not touch exposed connections and components when power is present.

Do Not Operate in Wet/Damp Conditions.

Do Not Operate in an Explosive Atmosphere.

Keep Product Surfaces Clean and Dry.

Provide Proper Ventilation

Refer to the manual's installation instructions for details on installing the product so it has proper ventilation.

Safety Requirements

This section contains information and warnings that must be observed to keep the instrument operating in a correct and safe condition. You are required to follow generally accepted safety procedures in addition to the safety precautions specified in this section.

Safety Symbols

Where the following symbols appear on the instrument's front or rear panels, or in this manual, they alert you to important safety considerations.



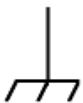
This symbol is used where caution is required. Refer to the accompanying information or documents in order to protect against personal injury or damage to the instrument.



This symbol warns of a potential risk of shock hazard.



This symbol is used to denote the measurement ground connection.



This symbol is used to denote a frame or chassis connection.



This symbol is used to denote a safety ground connection.



On (Supply). This is the DC power connect/disconnect switch at the back of the instrument.



Off (Supply). This is the DC power connect/disconnect switch at the back of the instrument.



This symbol is used to denote Power. It is located on the front panel and denotes Power On/Off status of the instrument.



This symbol is used to denote Direct Current.

CAUTION

The **CAUTION** sign indicates a potential hazard. It calls attention to a procedure, practice or condition which, if not followed, could possibly cause damage to equipment. If a **CAUTION** is indicated, do not proceed until its conditions are fully understood and met.

WARNING

The **WARNING** sign indicates a potential hazard. It calls attention to a procedure, practice or condition which, if not followed, could possibly cause bodily injury or death. If a **WARNING** is indicated, do not proceed until its conditions are fully understood and met.

CAT I

Installation (Overvoltage) Category rating per EN 61010-1 safety standard and is applicable for the instrument front panel measuring terminals. CAT I rated terminals must only be connected to source circuits in which measures are taken to limit transient voltages to an appropriately low level.

Operating Environment

The instrument is intended for indoor use and should be operated in a clean, dry environment. Before using this product, ensure that its operating environment is maintained within these parameters:

Temperature: 5 °C to 40 °C.

Humidity (operating): 5 % to 80 % relative humidity with a maximum wet bulb temperature of 29 °C at or below +40 °C, (upper limit de-rates to 20.6 % relative humidity at +40 °C . Non-condensing.

Humidity (non-operating): 5 % to 95 % relative humidity with a maximum wet bulb temperature of 40 °C at or below +60 °C, (upper limit de-rates to 29.8 % relative humidity at +60 °C. Non-condensing.

Altitude (operating): Up to 3,000 meters (9,842 feet) at or below 25 °C.

AC Power Source

For External AC Adapter: 100 to 240 VAC (+/-10%) at 45-66 Hz; Automatic AC voltage selection; Installation Category: 300V CAT II

No manual voltage selection is required because the external AC Adapter automatically adapts to line voltage.

Power Consumption

≤ 150 watts



WARNING - Electrical Shock Hazard

Only use the power cord provided with your instrument.

Calibration

The recommended calibration interval is one year. Calibration should be performed by qualified personnel only.

Cleaning

Clean only the exterior of the instrument, using a damp, soft cloth. Do not use chemicals or abrasive elements. Under no circumstances allow moisture to penetrate the instrument.

Abnormal Conditions

Operate the instrument only as intended by the manufacturer.

If you suspect the instrument's protection has been impaired, disconnect the power cord and secure the instrument against any unintended operation.

The instrument's protection is likely to be impaired if, for example, the instrument shows visible damage or has been subjected to severe transport stresses.

Proper use of the instrument depends on careful reading of all instructions and labels.



WARNING

Any use of the instrument in a manner not specified by the manufacturer may impair the instrument's safety protection.

Environmental Considerations

This section provides information about the environmental impact of the product.

Product End-of-life Handling

Observe the following guidelines when recycling an instrument or component.

Equipment Recycling. Production of this equipment required the extraction and use of natural resources. The equipment may contain substances that could be harmful to the environment or human health if improperly handled at the product's end of life. In order to avoid release of such substances into the environment and to reduce the use of natural resources, we encourage you to recycle this product in an appropriate system that will ensure that most of the materials are reused or recycled appropriately.



The symbol shown to the left indicates that this product complies with the European Union's requirements according to Directive 2002/96/EC on waste electrical and electronic equipment (WEEE).

Operating Requirements

Power Supply

Source Voltage and Frequency	100 to 240 V RMS@ 50-60Hz					
	Characteristic	condition	Min.	Nom.	Max.	Units
	Voltage	45-66Hertz	85	100-240	264	VRMS
		360-440 Hertz	100	115	132	VRMS
	Voltage Wave	All	Sine			
Power Consumption	Maximum: 150W Measured: 125W					
Surge Current	30 A peak (25°C) for 5 line cycles, after product has been turned off for at least 30 s.					

Mechanical Characteristics

Net Weight	10.5 kg
Net Weight with Package	11.5 kg
Overall Dimensions	Height: 160 mm
	Width: 450 mm
Dimensions with Package	Height: 210 mm
	Width: 500 mm

Environmental Characteristics

Temperature	Operating +5 °C to +50 °C (+41 °F to 122 °F)
Humidity	Operating 8% to 90% relative humidity with a maximum wet bulb temperature of 29 °C at or below +50 °C, non-condensing
	Non-operating 5% to 98% relative humidity with a maximum wet
Altitude	Operating 3,000 m (9,843 feet)
	Non-operating 12,000 m (39,370 feet)

Preface

This manual describes the installation and operation of Model 765 instruments Revision B. Basic operations and concepts are presented in this manual.

The following instruments are supported by this manual:

- Model 765-2 Rev.B
- Model 765-4 Rev.B

The Model 765 Series offers premium signal integrity with the easiest to use touch screen display interface (SimpleRider™). The Generation of pulses requires only a few screen touches.

The output Voltage can be adjusted up to 5 Volts pk-pk in a window of ± 5 Volts with 70 ps edge rate (based on RiderEdge™ technology) and transitions with minimal overshoot and ringing.

Its innovative hardware architecture provides the possibility to generate advanced pulse sequences, such as double pulse or quad pulse, with fully independent timing parameters.

Package Contents

The standard Model 765 package includes the following:

- Model 765-2 or Model 765-4 pulse generator instrument
- Power Cord
- Performance/Calibration Certificate
- Model 765 Introduction and Compliance document

2-4 Channels

Pulse Generator is always available with the basic Dual Channel (765-2) version or with the Quad Channel version (765-4).

Each channel may generate pulses with rise time as low as 70 ps, thanks to the **RiderEdge™** amplifier, and frequency repetition rate from mHz to 125 MHz in single pulse mode and 800 MHz in quadruple pulse mode.

By using the Multiple output pulses feature, it is also possible to reach a max. repetition rate of 800 MHz.

Output Voltage is fully adjustable up to 5 Volts pk- pk inside a voltage window of ± 5 Volts.

The offset is adjustable inside a voltage window of ± 2.5 Volts.

The new family of **RiderPulse™** Generator can produce Multiple output pulses (i.e. double pulse, quad pulse) with independent repetition rate, width, delay, amplitude and polarity.

This gives the possibility to use the instrument as a digital delay generator for rescaling, synchronizing, delaying, gating and triggering multiple devices with respect to one unique event.

Trigger, view, generate and sync

Trigger events may be generated internally or captured by an external trigger source or remotely from Ethernet or GPIB connections.

Trigger In and Trigger Out may be used to synchronize multiple units to obtain several pulses and to provide a perfect solution for specific Big Physics, Military applications or Semiconductor testing.

Simple Rider UI is designed for touch and it has been developed to put all the capabilities of the modern Pulse and Waveform Generators right at your fingertips.

Touch Screen display and Soft Keyboard

The new Rider Series delivers 7" capacitive touch screen display and the touch-screen friendly SimpleRider™ software allows users to generate pulses quickly by a few screen touches.

The UI ergonomic approach is well balanced to offer multiple ways to operate the instrument by offering a complementary soft keyboard and a useful central knob for fine-tuning and adjustments during the set up operation.

SimpleRider Pulse Touch User Interface

All instrument controls and parameters are accessed through an intuitive UI that recalls the simplicity of Tablets and modern smart phones: touch features and gestures are available to engineers and scientists to create pulses or complex combination of pulses in few touches.

The swipe gesture gives easy access to the output and pulse parameters.

- A touch-friendly virtual numeric keypad has been designed to improve the user experience on entering the data.
- Time saving shortcuts and intuitive icons simplify your setup also during pulse combination operations.
- Model 765 supports the most common interfaces for remote control (Ethernet, GPIB) for easy customized instrument programming

Recommended

Accessories

Item	Description
RM KIT	Rack Mount Kit
SSD KIT	SSD Solid State Drive
WAR EXT	Warranty Extension
CAL. CERT	Certification of Calibration

Power the Instrument On and Off

Power On

- Insert the AC power cord into the power receptacle on the rear panel.
- Use the front-panel power button to power on the instrument.
- Wait until the system shows windows desktop.
- The Model 765 software will start automatically

Power Off

- Close the application in use.



- Press the front-panel power button  to power off the instrument.



Protect Your Instrument from Misuse

Check Input and Output Connectors

When connecting a cable, be sure to distinguish the input connector from the output connectors to avoid making the wrong connection.



The instrument has both input and output connectors on the front panel. When connecting a cable, be sure to distinguish the input connectors from the output connectors.

CAUTION. Do not short output pins or apply external voltages to Output connectors. The instrument may be damaged.

CAUTION. Do not apply excessive inputs over $\pm 15\text{Vpk}$ to Trigger Input connector. The instrument may be damaged.

Obtaining the Latest Version Releases

The latest version of an optional application that you ordered with your instrument may not be installed on your instrument. The following download location is a fast and easy way to get the latest software version.

To download the latest version of software, register on the website; go to the home page of the Berkeley Nucleonics website (www.berkeleynucleonics.com), press the Register button in the upper right of your screen.



Then navigate to the Product page and scroll down to the Software, Firmware and Drivers section. Then simply click on the link to start the download.

SOFTWARE, FIRMWARE AND DRIVERS

SDK-AT-Pulse Generator Software

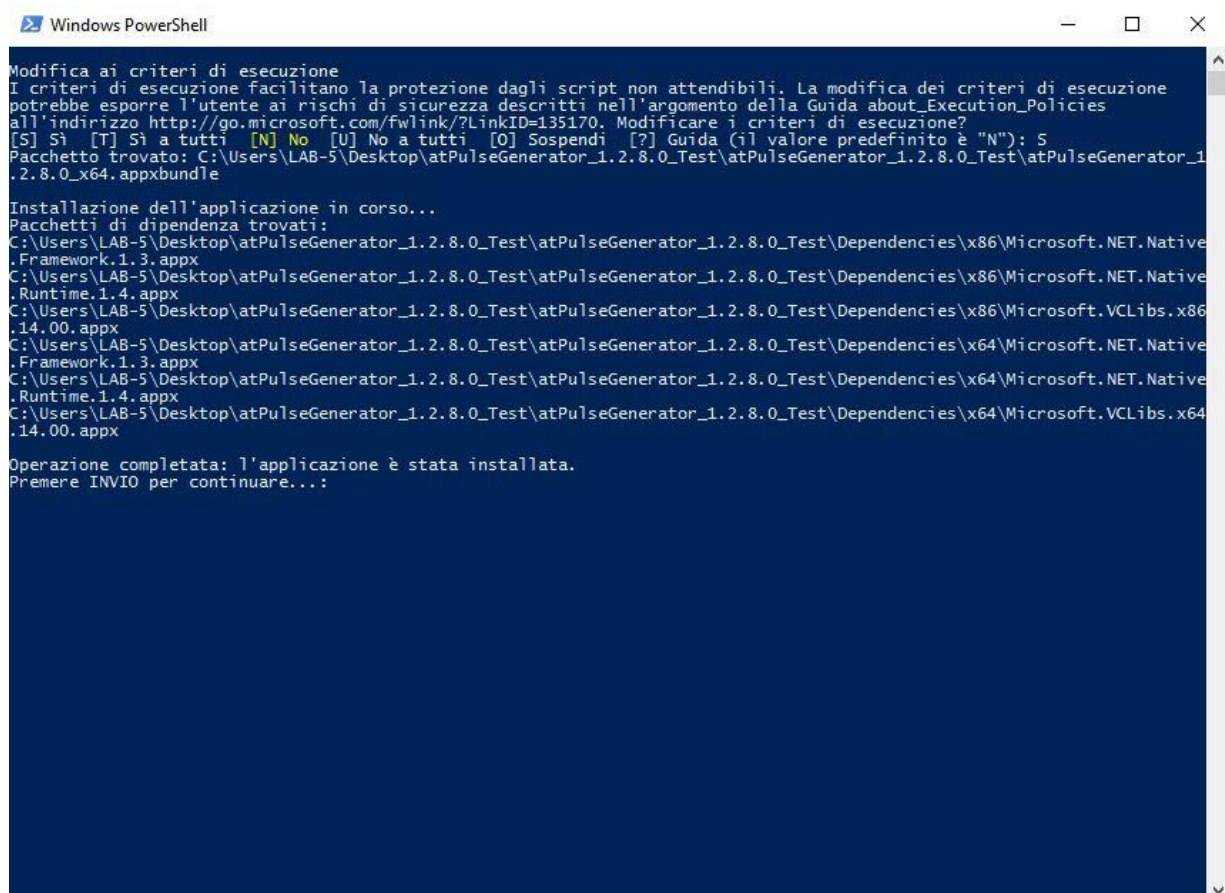
Install the Model 765 Pulse Generator Application

If your instrument has already installed another version of the Pulse Generator APP, you must first uninstall it.

- 1. Download the Model 765 Pulse Generator setup package from the Berkeley Nucleonics website and decompress it to instrument’s local disk.

Nome	Ultima modifica	Tipo	Dimensione
Add-AppDevPackage.resources	30/03/2017 18:25	Cartella di file	
Dependencies	30/03/2017 18:25	Cartella di file	
Add-AppDevPackage.ps1	07/07/2015 01:51	Script di Windows...	61 KB
atPulseGenerator_1.2.8.0_x64.appxbundle	30/03/2017 18:25	File APPXBUNDLE	13.062 KB
atPulseGenerator_1.2.8.0_x64.appxsym	30/03/2017 18:24	File APPXSYM	13.018 KB
atPulseGenerator_1.2.8.0_x64.cer	30/03/2017 18:25	Certificato di sicur...	2 KB

2. Right click on the Add-AppDevPackage.ps1 file and select "Run with PowerShell"
3. Follow the steps on the script



```
Windows PowerShell

Modifica ai criteri di esecuzione
I criteri di esecuzione facilitano la protezione dagli script non attendibili. La modifica dei criteri di esecuzione
potrebbe esporre l'utente ai rischi di sicurezza descritti nell'argomento della Guida about_Execution_Policies
all'indirizzo http://go.microsoft.com/fwlink/?LinkID=135170. Modificare i criteri di esecuzione?
[S] Sì [T] Sì a tutti [N] No [U] No a tutti [O] Sospendi [?] Guida (il valore predefinito è "N"): S
Pacchetto trovato: C:\Users\LAB-5\Desktop\atPulseGenerator_1.2.8.0_Test\atPulseGenerator_1.2.8.0_Test\atPulseGenerator_1
.2.8.0_x64.appxbundle

Installazione dell'applicazione in corso...
Pacchetti di dipendenza trovati:
C:\Users\LAB-5\Desktop\atPulseGenerator_1.2.8.0_Test\atPulseGenerator_1.2.8.0_Test\Dependencies\x86\Microsoft.NET.Native
.Framework.1.3.appx
C:\Users\LAB-5\Desktop\atPulseGenerator_1.2.8.0_Test\atPulseGenerator_1.2.8.0_Test\Dependencies\x86\Microsoft.NET.Native
.Runtime.1.4.appx
C:\Users\LAB-5\Desktop\atPulseGenerator_1.2.8.0_Test\atPulseGenerator_1.2.8.0_Test\Dependencies\x86\Microsoft.VCLibs.x86
.14.00.appx
C:\Users\LAB-5\Desktop\atPulseGenerator_1.2.8.0_Test\atPulseGenerator_1.2.8.0_Test\Dependencies\x64\Microsoft.NET.Native
.Framework.1.3.appx
C:\Users\LAB-5\Desktop\atPulseGenerator_1.2.8.0_Test\atPulseGenerator_1.2.8.0_Test\Dependencies\x64\Microsoft.NET.Native
.Runtime.1.4.appx
C:\Users\LAB-5\Desktop\atPulseGenerator_1.2.8.0_Test\atPulseGenerator_1.2.8.0_Test\Dependencies\x64\Microsoft.VCLibs.x64
.14.00.appx

Operazione completata: l'applicazione è stata installata.
Premere INVIO per continuare...
```

4. When the application has been installed, press Return to continue.

Remote Control

You can connect your instrument to a network for printing, file sharing, and Internet access, among other functions. Consult with your network administrator and use the standard Windows utilities to configure the instrument for your network.

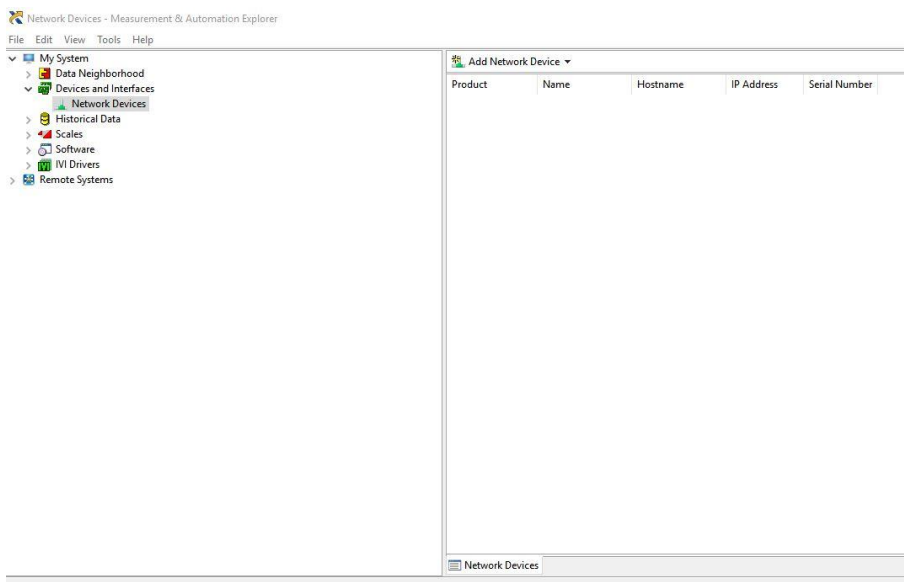
The instrument can be controlled using VXI-11 (LAN) protocol. It allows you to control the instrument remotely by using SCPI commands. Please refer to the Model 765 programmer manual for a complete description about all available commands.

You can follow the next steps to communicate with your Model 765 Series instrument.

NI-VISA

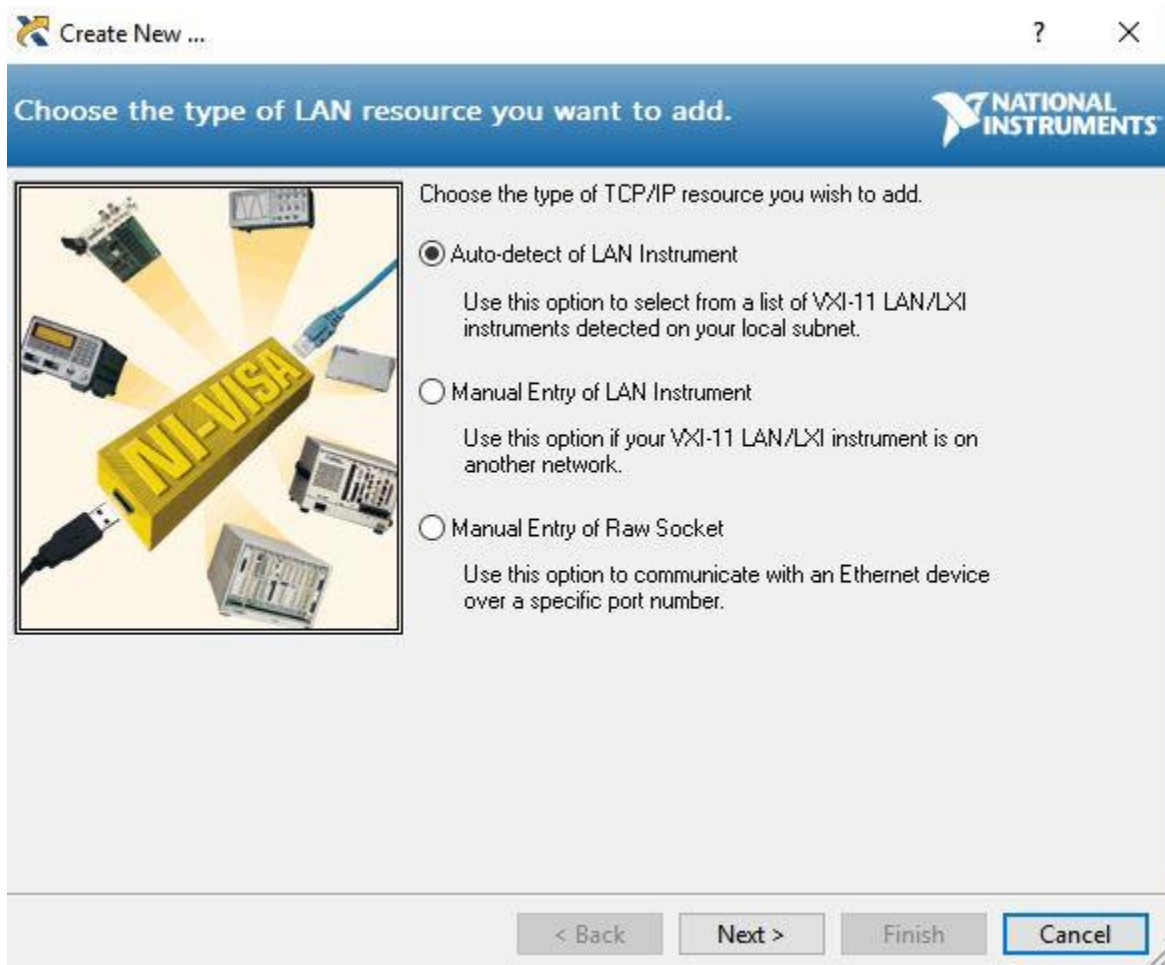
VISA provides the programming interface between the hardware and development environments such as Visual Studio .NET, LabVIEW, LabWindows/CVI, Measurement Studio for Microsoft Visual Studio and MatLab. NI-VISA is the National Instruments implementation of the VISA I/O standard. NI-VISA includes software libraries, interactive utilities such as NI I/O Trace and the VISA Interactive Control, and configuration programs through Measurement & Automation Explorer for all your development needs.

1. Connect your LAN cable to the instrument.
2. On the Client-PC you must install the latest NIVISA package that you can find here
<http://search.ni.com/nisearch/app/main/p/bot/no/ap/tech/lang/it/pg/1/sn/catnav:du,n8:3.1637,ssnav:sup/>
3. Launch the NI-MAX tool on the Client-PC

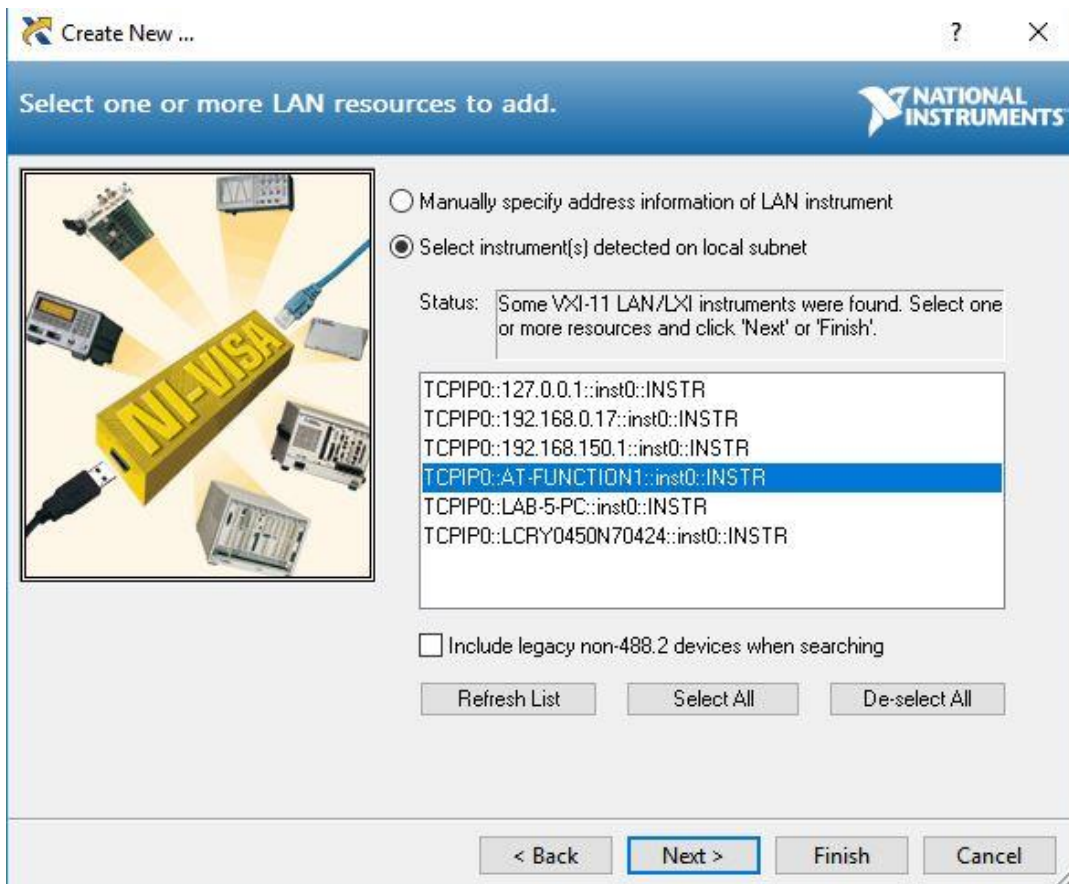


Press Add Network Device → VISA TCP/IP Resource...

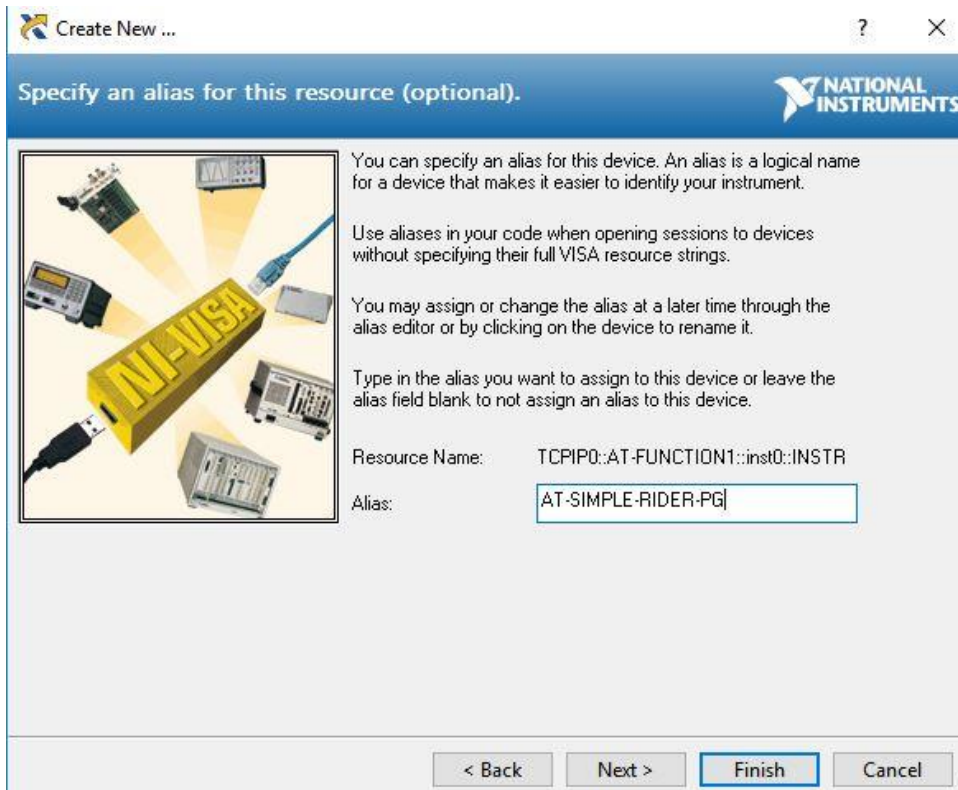
4. Select Auto-detect of LAN Instrument



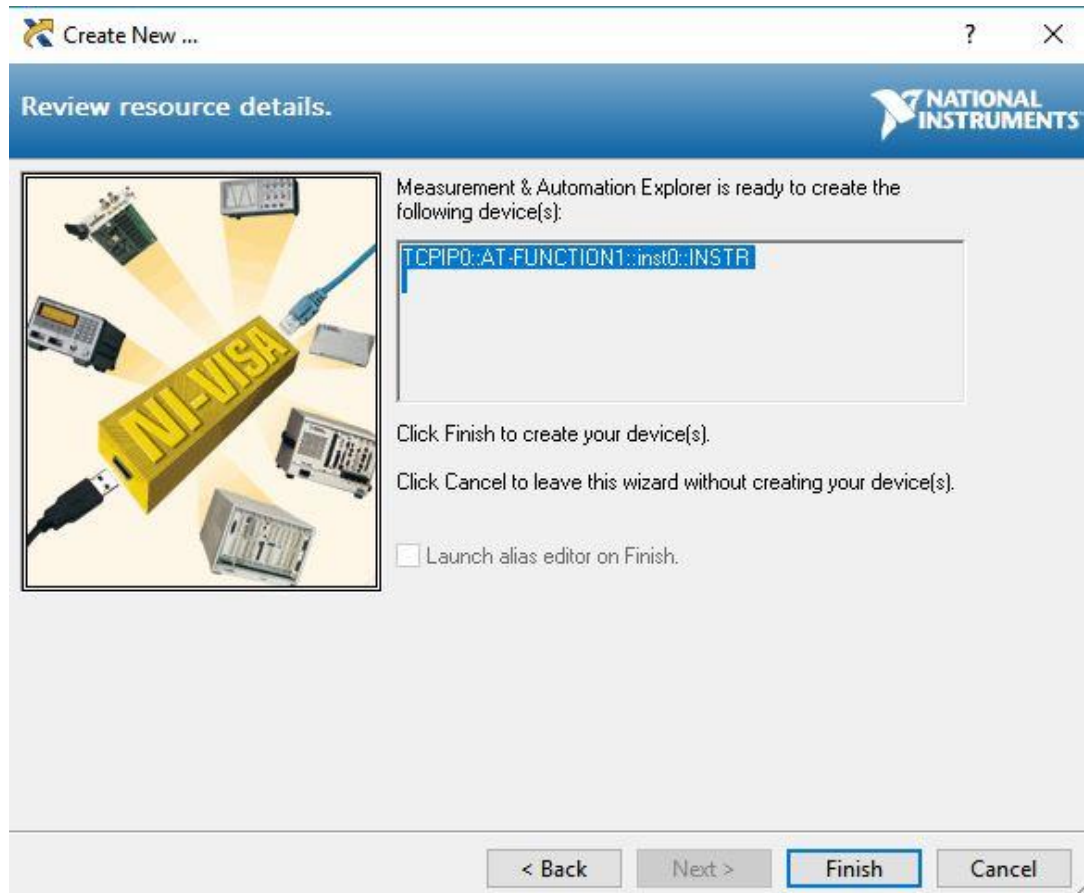
5. The panel will retrieve the discovered instruments on the LAN network, you should select the Model 765 series one.



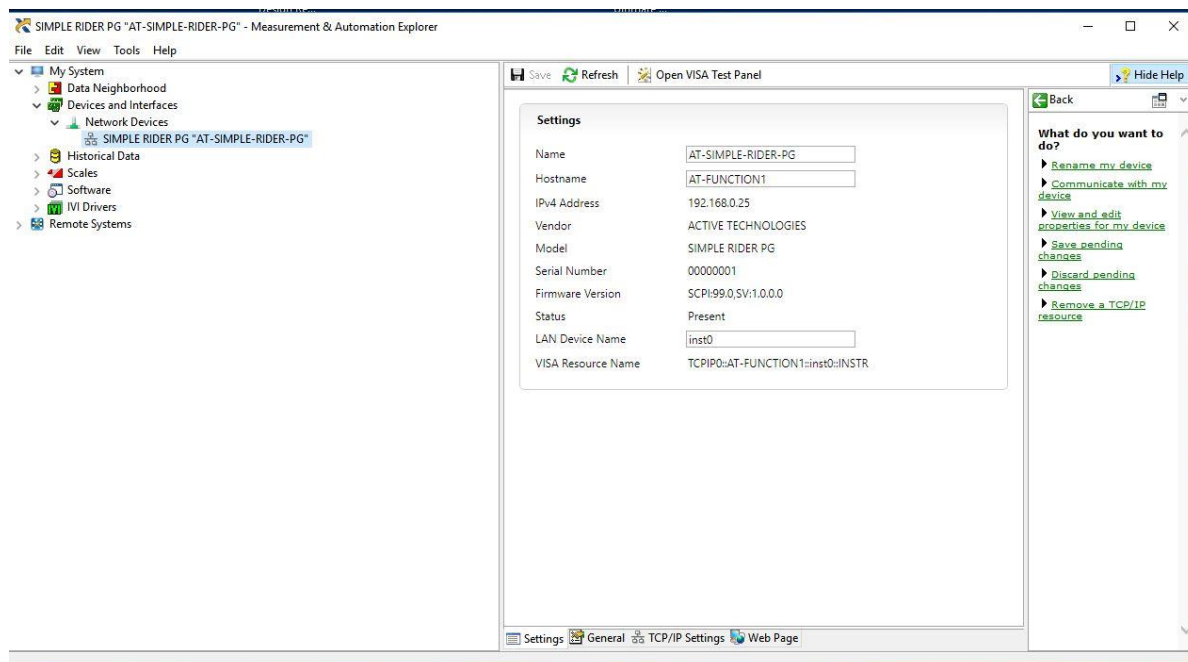
6. Specify an Alias for the selected resource



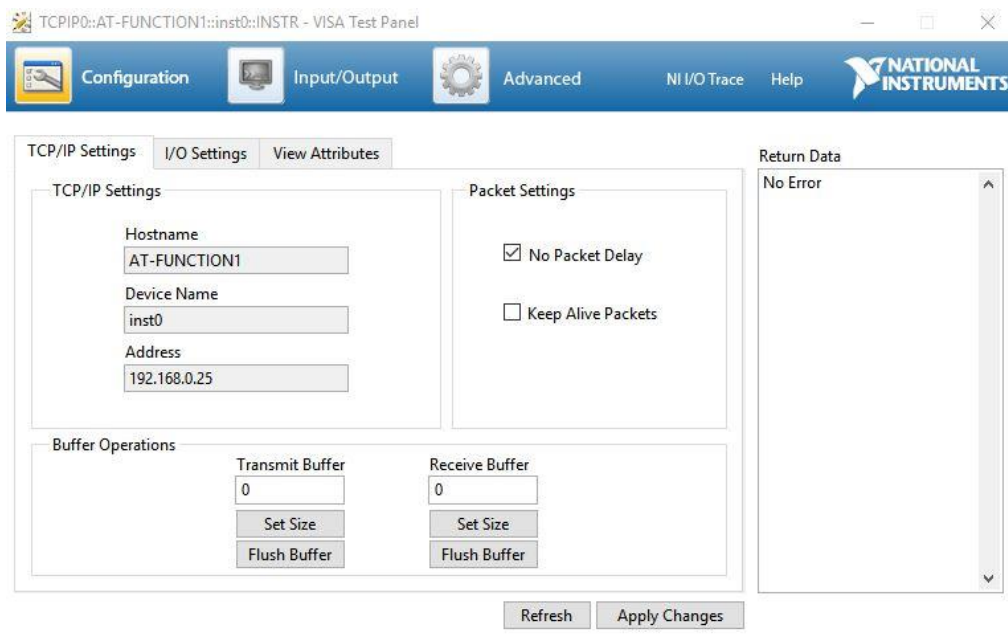
7. Press Finish



8. The Model 765 resource will be available in the Network Devices list

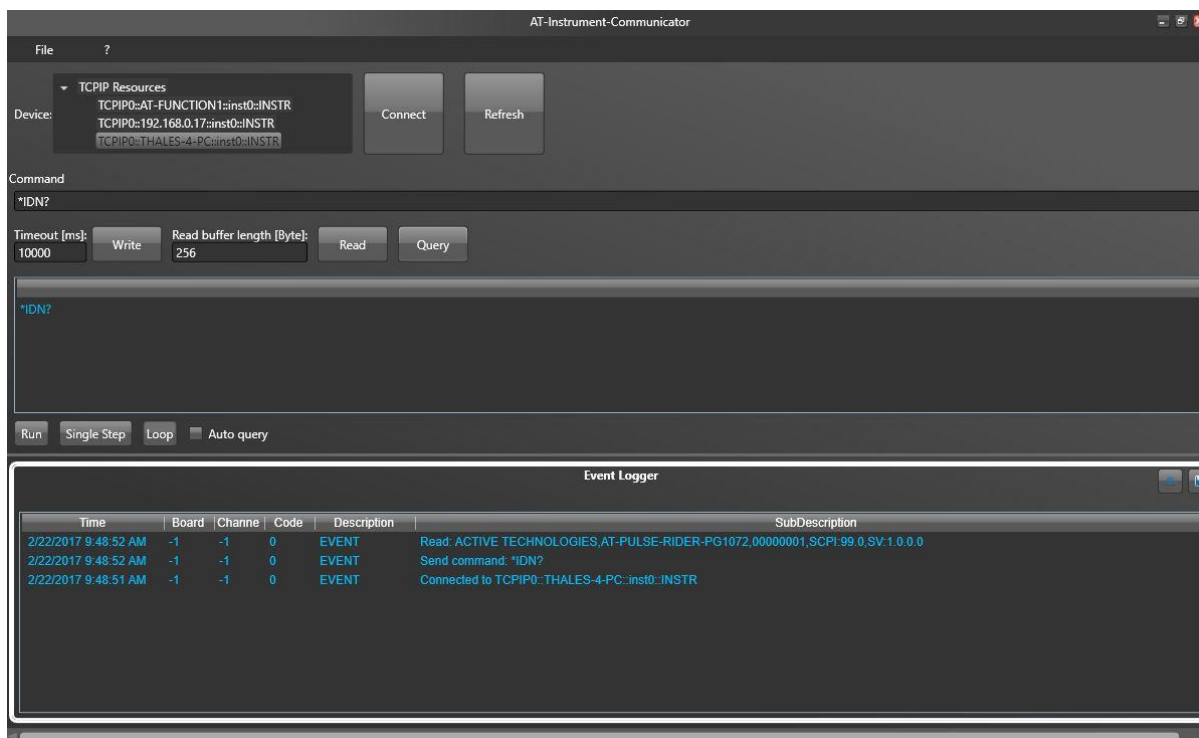


9. Now you can use send the SCPI commands to the Model 765 resource using the NI Visa Test Panel or the AT-Instrument-Communicator



- On the Client-PC (IP Address) or Model 765 instrument (Local Host), launch the AT-Instrument-Communicator tool

AT Instrument Communicator



The instrument VXI-11 LAN Server provides software connectivity between your instrument and remote PCs over an Ethernet LAN.

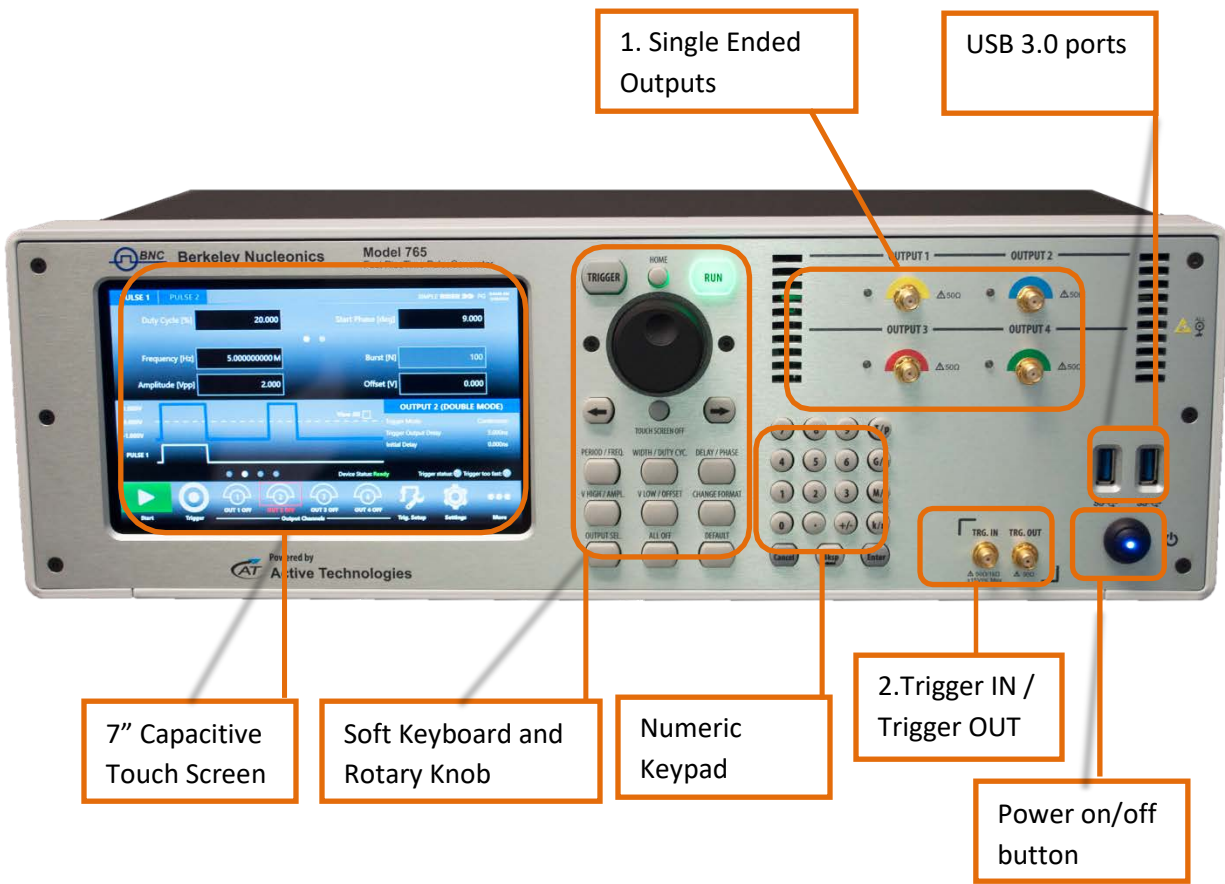
The AT-Instrument-Communicator software is a client-side component tool that uses NI-VISA on each remote PC, you must install a copy of NIVISA to make use of this client-side component (please follow the Prerequisite steps).

1. On the Client-PC launch the *AT Instrument Communicator* setup and install the software.
2. Select the Model 765 resource on the Device list
3. Press the Connect button
4. If the instrument connection will be established, the SCPI command button will be enabled.
5. Write *IDN? in the command
6. Press the Query button
7. In the Event Logger list, the instrument should respond like this:
ACTIVE TECHNOLOGIES, AT-PULSE-RIDER PG1072 00000001,SCPI 99.0, SV 1.0.0 where 00000001 is the serial number, SCPI 99.0 is the SCPI command version and SV 1.0.0 is the software version.
8. A command script is a list of SCPI commands (one command for each line) saved in a txt file; you can send a command script using the File → Load Script menu item.
9. Refer to the SCPI programmer manual for a complete list of all available commands.

Instrument Overview

Front Panel

The numbered callouts on this image correspond with the following descriptions.



1. ANALOG OUTPUTS

OUTPUT 1 – Single Ended DC Coupled Analog output 1 (SMA connector).

OUTPUT 2 – Single Ended DC Coupled Analog output 2 (SMA connector).

OUTPUT 3 – Single Ended DC Coupled Analog output 3 (SMA connector) – PG1074 only.

OUTPUT 4 – Single Ended DC Coupled Analog output 4 (SMA connector) – PG1074 only.

The DC coupled is a single output with 50 Ohm output impedance each; in case you need a differential output you should use the OUTPUT 1 and the OUTPUT 2 setting

them in the SimpleRider UI as positive and negative pulses.

2. TRIGGER INPUT/OUTPUT

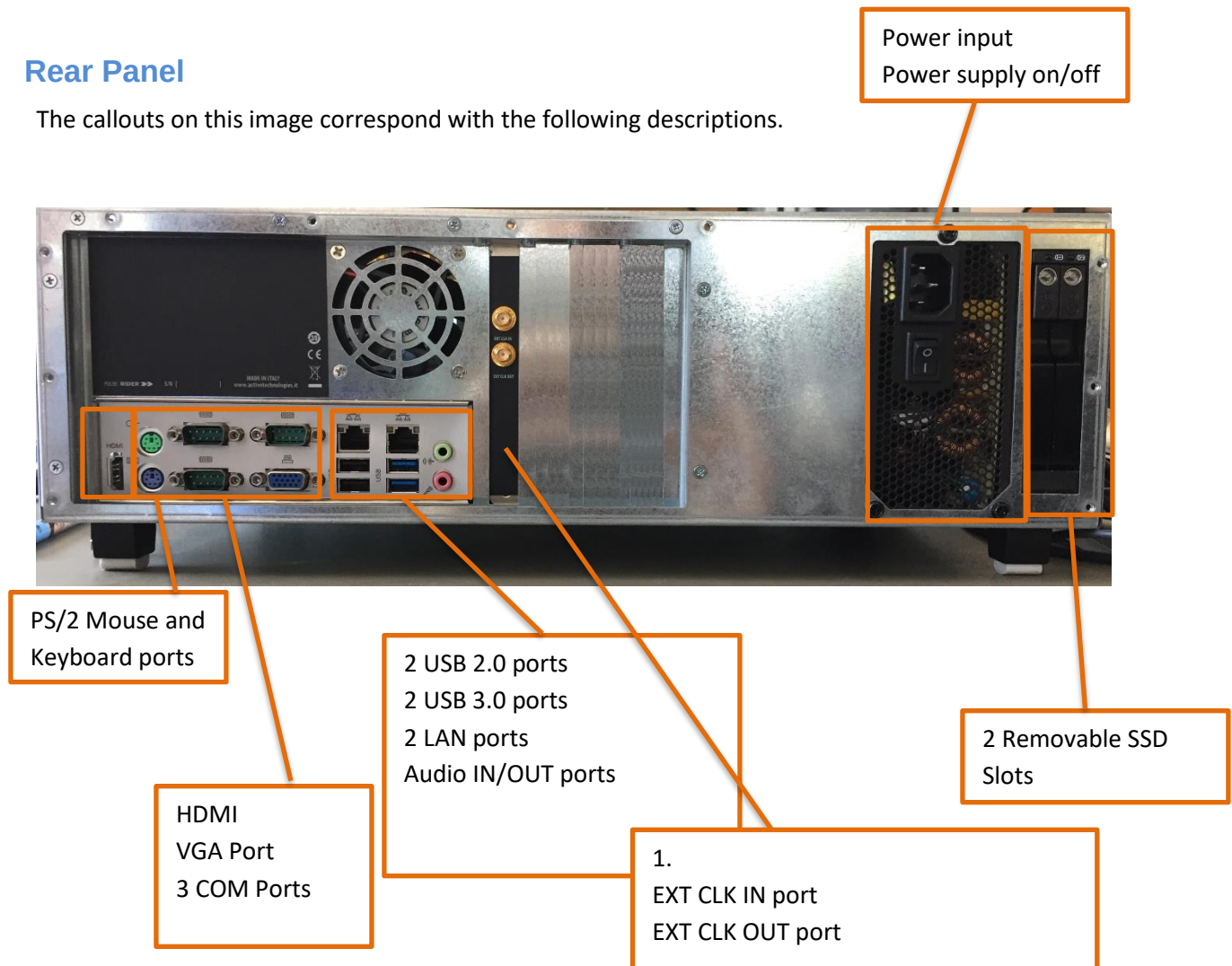
TRG.IN - SMA input connector for Trigger IN.

The **Trigger In** input has a programmable impedance of 50 Ohm or 1k Ohm and a programmable threshold level in the range -10V to 10V.

TRG.OUT - SMA output connector for Trigger OUT. The Trigger output impedance is 50 Ohm and the voltage range on open load is from 1.8V to 3.3V.

Rear Panel

The callouts on this image correspond with the following descriptions.



1. **EXT. CLK IN Port** - SMA Input for External Clock IN.

The Model 765 can use an external clock source to generate the instrument time base. This feature allows to synchronize the generator with an external clock.

EXT. CLK OUT Port – SMA Output for External Clock OUT.

If the clock source is internal this connector outputs the internal 10Mhz reference clock, if the source is external this connector outputs the Ext.Clk.IN value.

Getting Started with Model 765

Instrument Control

The Model 765 instrument includes a 7" capacitive touch screen and Simple Rider touch user interface based on a Microsoft Windows platform.

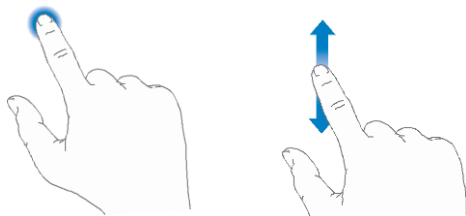
You can control instrument operations using the following:

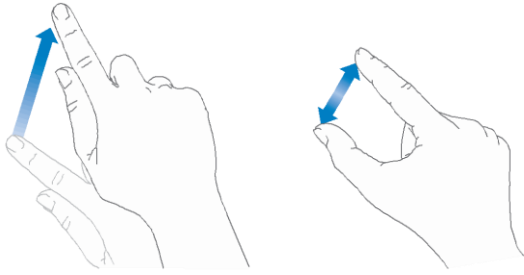
- Touchscreen
- Front-panel soft key controls
- Keyboard and mouse

SimpleRider Touch UI

SimpleRider UI is designed for touch to drive simplicity in operating with a pulse generator, by optimizing the today's modern technique, used on Tablet or smart phones, of capacitive touch screen display.

All important instrument controls and settings are always one touch away: swipe gesture to change the channel, pulse selection and have access to the modulation parameters, combine multiple pulses easily using the touch- combiner, use the touch-friendly virtual numeric keyboard to change parameter values on the fly.





Front Panel Buttons

Most of the buttons you use with Model 765 are virtual ones on the touchscreen, but a few physical buttons control basic functions, such as setting the pulse amplitude, offset, delay, width etc.

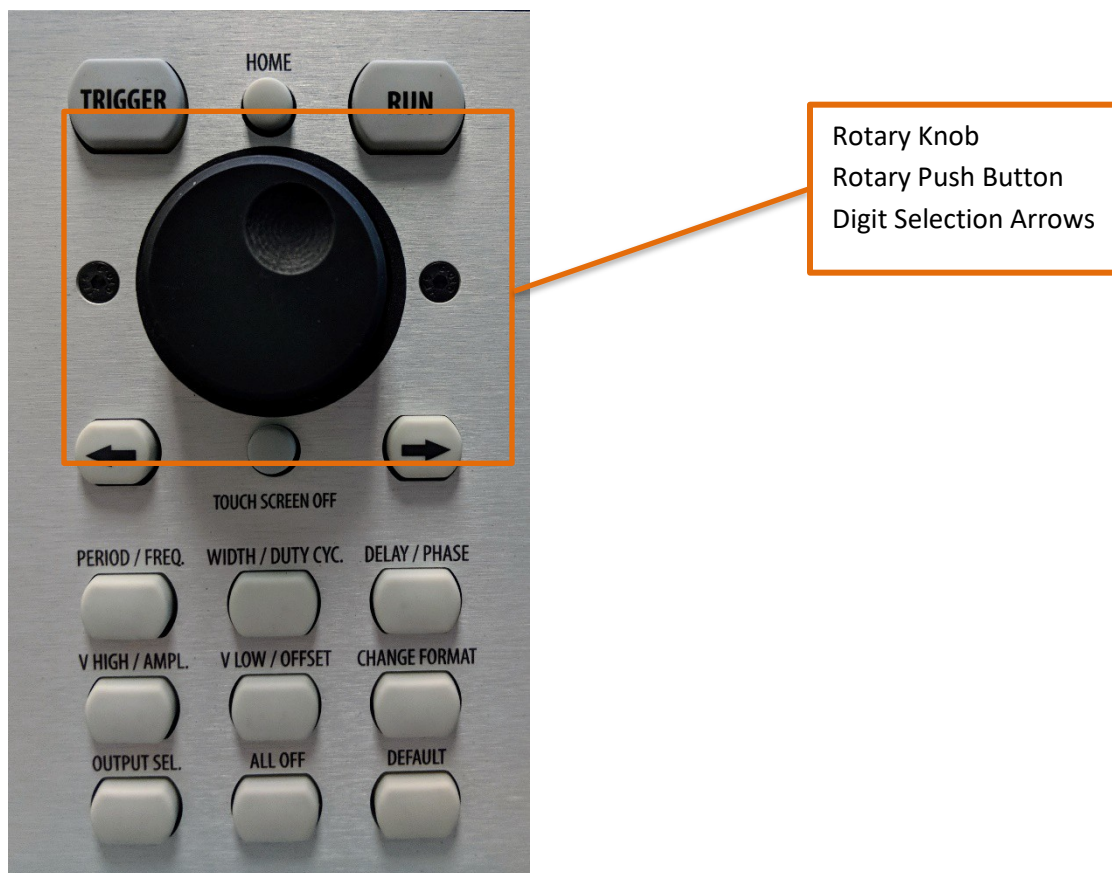
A physical numeric keypad is available on the front-panel and it can be used instead of the virtual numeric pad.

A useful central knob is available for fine-tuning and adjustments during the on the fly set up operation.

The rotary knob will change the value in continuous, analog fashion. The push button rotary knob lets you to change the value increment between Coarse and Fine adjustment.

The → key will move the selected digit to the right and the ← key will move the selected digit to right.

You can keep pressed the rotating knob and rotate it on the right or on the left to change the Delta increment.



Button Name	Function
HOME	If you are in a software sub-menu page, use this button to return to the main page.
TRIGGER	Use this button to send an internal trigger to the instrument.
RUN	Use this button to start and stop the pulse generation. When the button led is green, it means that the instrument is in running state (or it is armed, ready to receive the trigger signal).
LEFT ARROW	Once the virtual numeric keypad will be opened, use this button to move left the digit selection cursor.
RIGHT ARROW	Once the virtual numeric keypad will be opened, use this button to move right the digit selection cursor.
TOUCH SCREEN OFF	Use this button to disable the touch screen.

PERIOD/FREQ	Use this button to set the period or the frequency of the pulse waveform.
WIDTH / DUTY CYC.	Use this button to set the width or the duty cycle of the pulse waveform.
DELAY/PHASE	Use this button to set the delay or the phase of the pulse waveform.
V HIGH/AMPL.	Use this button to set the high voltage level or the amplitude of the pulse waveform.
V LOW/OFFSET	Use this button to set the low voltage level or the offset of the pulse waveform.
CHANGE FORMAT	Use this button to change the format of the data entry between Format A and Format B. Format A is: Period, Width, Delay, Phase, Voltage Level High, Voltage Level Low. Format B is: Frequency, Duty Cycle, Phase, Amplitude, Offset. Some pulse parameters can be controlled in alternate formats. Period for example, displays and controls the time interval between pulses. If you press this button while Period is selected, the display will change to Frequency Now, you can control and read back the frequency at which pulses are output.
OUTPUT SEL.	Use this button to change the output selection in the user interface
ALL OFF	Use this button to turn off all the outputs.
DEFAULT	Use this button to restore the default settings.

Front Panel Numeric Keypad

The physical numeric keypad lets you to set the pulse parameter value and their measure unit.

Use the touch panel to select the pulse parameter, each number pressed will be displayed in the display, the Bksp key is provided for deleting erroneous key presses. The [+/-] key will toggle the sign of the number being entered and may be pressed at any time before terminating entry. After the sign and numeric portion of the desired value have been punched in, entry with the appropriate multiplier is terminated by pressing one of the four unit measure keys at the right of the keypad.



When you select a parameter on the user interface, if you press the unit measure button it will automatically set the available range linked to the parameter.

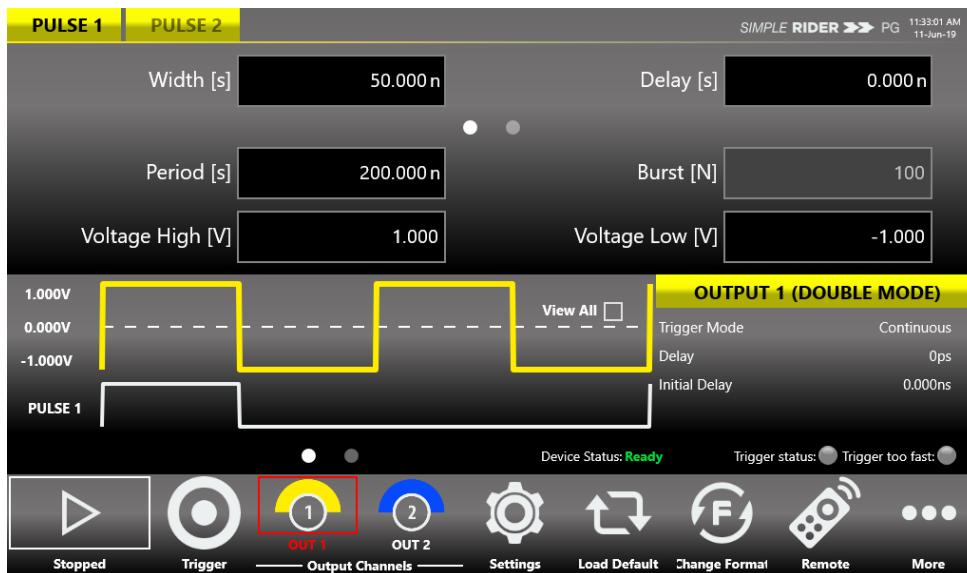
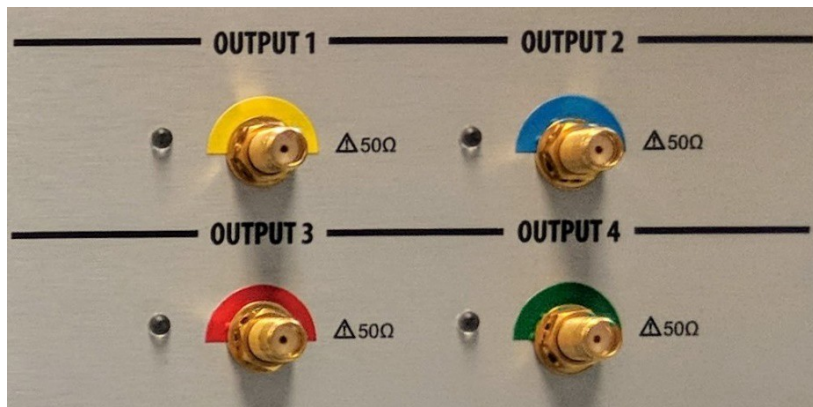
Unit Measure Range Button	Unit Measure Range
T/p	Tera / pico
G/n	Giga / nano
M/u	Mega / micro
k/m	Kilo / milli

As example when you select a Frequency parameter and you press k/m the unit measure range that will be inserted is kHz, if you press M/u it will be inserted MHz, if you press M/u it will be inserted GHz, if you press T/p nothing will happen because that range is not available for the selected parameter.

If more than one unit measure range is available for a selected parameter (i.e. Mega and Micro), if you toggle the range button **M/u**, the range will change accordingly between Mega and Micro.

Output and Pulse Definition

The term Output Channel as used in this user manual, refers to the analog signal at the SMA output connectors (labels OUTPUT 1, OUTPUT 2, OUTPUT 3, OUTPUT4) on the front panel.



The Output Channel parameters are:

- Voltage High / Amplitude [V]
- Voltage Low / Offset [V]

- Period [s] / Frequency [Hz]

The term Pulse refers to the logical pulse that it is internally generated by the digital core of the Model 765 instrument and that it is used to compose the analog Output channel signal.

The screenshot shows the Simple Rider user interface with two main sections. The top section, labeled 'PULSE 1' and 'PULSE 2', contains 'Width [s]' set to 50.000 n and 'Delay [s]' set to 0.000 n. The bottom section contains 'Period [s]' set to 100.000 n, 'Burst [N]' set to 100, 'Voltage High [V]' set to 2.500, and 'Voltage Low [V]' set to -2.500. Two callout boxes on the right point to these sections: 'Pulse Parameters' points to the top section, and 'Output Channel Parameters' points to the bottom section.

Up to four digital pulses called PULSE1, PULSE2, PULSE3 and PULSE4 are available for each Output channel through the SimpleRider user interface; this feature lets you to generate Multiple Pulses at the output channel.

The Pulse parameters are:

- Width [s] / Duty Cycle[%]
- Delay [s] / Phase [degree]

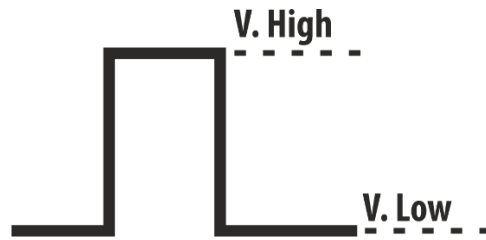
Output Channel Parameters

This screenshot is identical to the one above, showing the Simple Rider user interface with Pulse and Output Channel parameters. The 'Output Channel Parameters' section (bottom) is highlighted with an orange border, showing 'Period [s]' (100.000 n), 'Burst [N]' (100), 'Voltage High [V]' (2.500), and 'Voltage Low [V]' (-2.500).

Vertical Parameters

The Model 765 default format provides the control of the vertical (voltage) parameters of the Output channel.

Voltage High[V] controls the output high voltage level, **Voltage Low[V]** controls the output low voltage level.

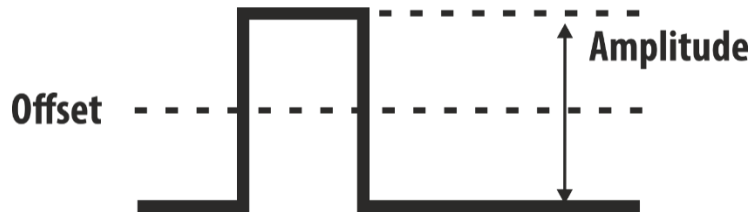


The output signal levels indicated by the SimpleRider UI will be delivered into 50 Ohm load.

Pressing the Change Format front panel or touch button will change the display of those parameters to Amplitude[V] and Offset[V].

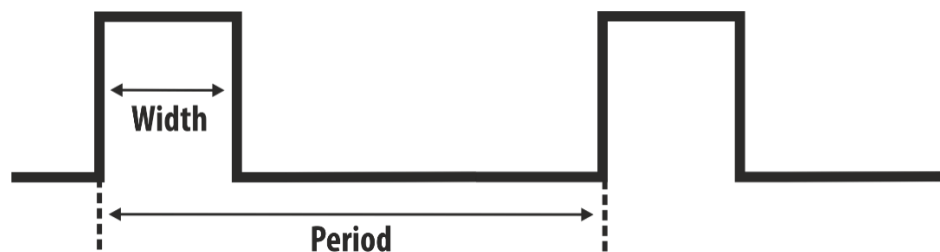
Amplitude[Vpp] is the difference between the active and the quiescent levels of the Output pulse.

Offset[V] is the quiescent level of the signal of the Output channel.



Period / Frequency

Period[s] parameter defines the time between Output pulses in Continuous, Burst and Gated mode.



Pressing the Change Format front panel or touch button will change the display of that parameter to Frequency.

Frequency[Hz] controls the frequency of Output pulses in Continuous, Burst and Gated mode.

Pulse Parameters

PULSE 1		PULSE 2		SIMPLE RIDER PG 3/4754 Rev 3/1/2017	
Width [s]	50.000 n	Delay [s]	0.000 n		
Period [s]	100.000 n	Burst [N]	100		
Voltage High [V]	2.500	Voltage Low [V]	-2.500		

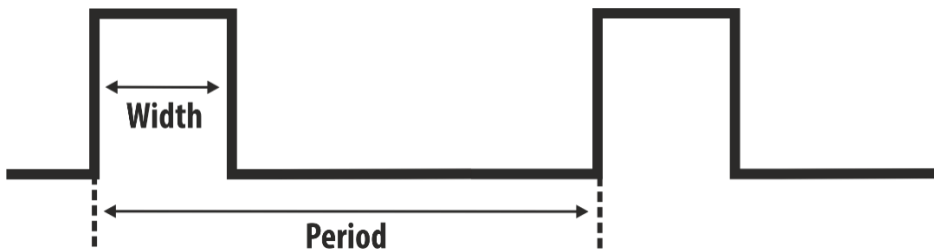
Width / Duty Cycle

The time duration of the Pulse is controlled by the Width / Duty Cycle parameter.

The **Width[s]** defines the time interval during the Pulse N is in the active state.

Pressing the Change Format button, the mode of control over the duration of the Pulse N will change to Duty Cycle.

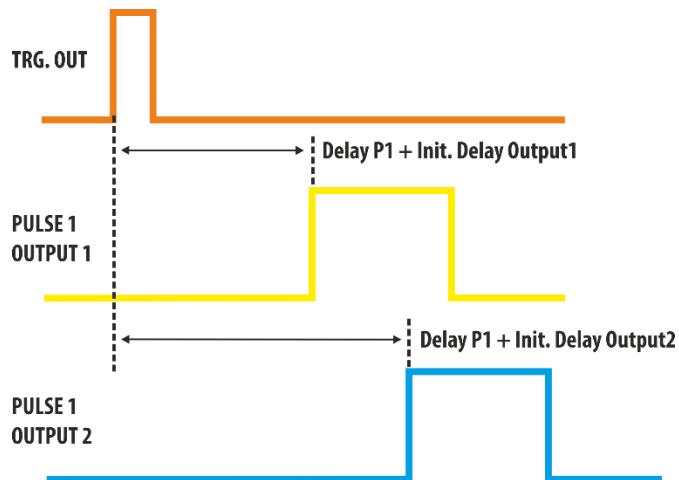
The **Duty Cycle[%]** defines the percentage of the Period over which the Pulse N is in the active state.



Delay / Phase

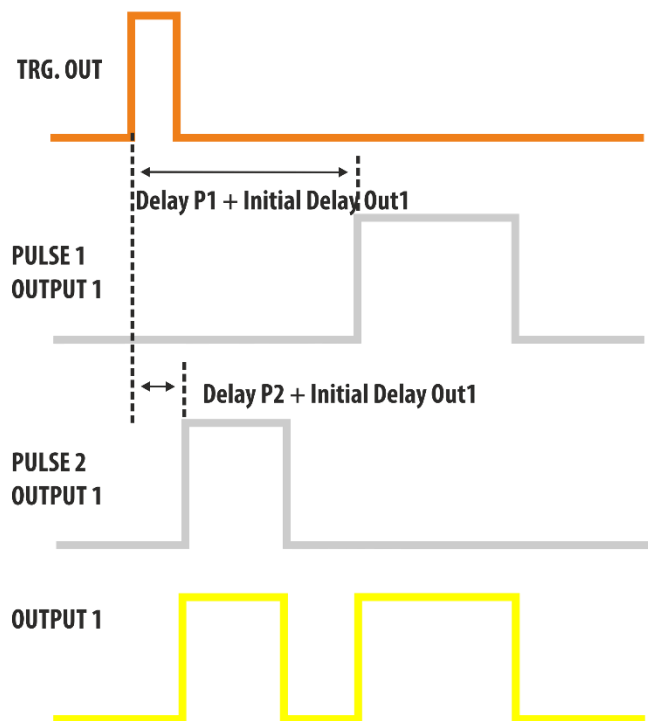
The Delay[s] parameter defines the time interval from the Trigger Output to the Pulse N, excluding an initial output channel delay that can be set in the Settings page.

In the figure below the Output 1 and Output 2 channels work in Single Pulse mode, it means that the Output signals is created by Pulse1/Output1 and Pulse1/Output2.



In the figure below the Output 1 channel works in Double Pulse mode, it means that the Output signals is created by the logical OR combination of Pulse1/Output1 and Pulse2/Output1.

DOUBLE PULSE MODE



If you press the Change Format button, the displayed name for the Delay parameter will change to Phase.

Phase[degree] provide position control in a manner proportional to Period, the Pulse N position is expressed as a phase angle, with 0 degree corresponding to the minimum Delay setting.

This phase angle is maintained as Period is varied. When Phase has been set, the Pulse Delay = $(\text{Phase} / 360) * \text{Period}$.

Parameter Conflicts

The following conditions must be true in order to avoid conflicts:

1. Voltage High > Voltage Low
2. Width[s]+Delay[s] < Period[s]

Parameter Limits

Parameter	Min	Max
Voltage High	-5.0 V	+5.0 V
Voltage Low	-5.0 V	+5.0 V
Amplitude	-2.5 V	+2.5 V
Offset	-2.5 V	+2.5 V
Width	300 ps	8 s
Duty Cycle	1 %	99.9%
Period	5 ns	8 sec
Frequency	0.125 Hz	200 MHz
Delay	0 s	8 sec
Phase	0 degree	360 degree
Burst N	1	4294967295

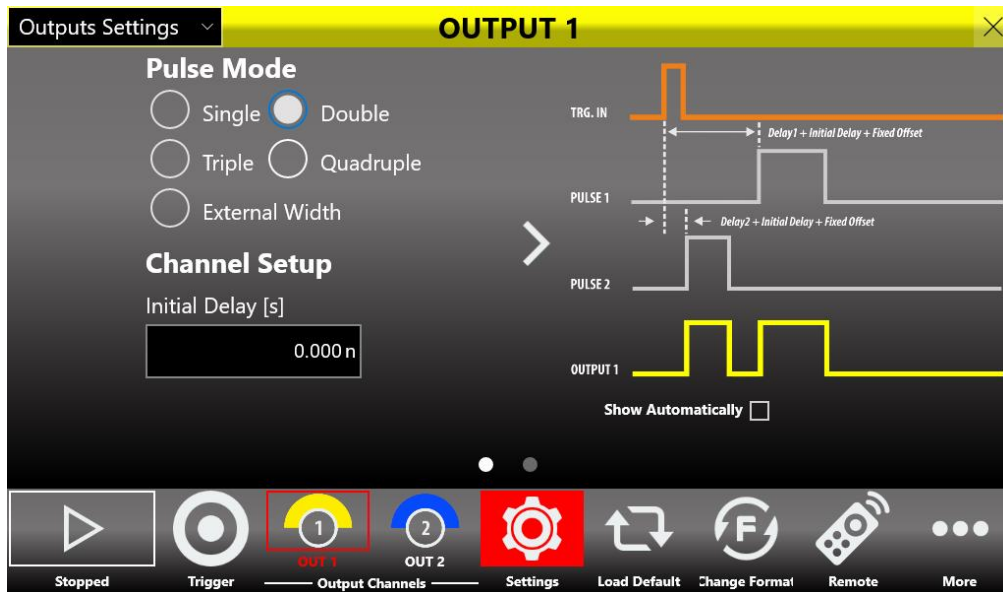
Multiple Pulses Operation

Four digital Pulses are available on each Output and they are logically combined by a OR operation to give you the possibility of generating Multiple Pulse on the output SMA connector reaching up to 800 MHz as repetition rate.

In Continuous and Single Trigger modes when the Multiple Pulse feature is turned off, one Output Pulse will follow each Trigger Output.

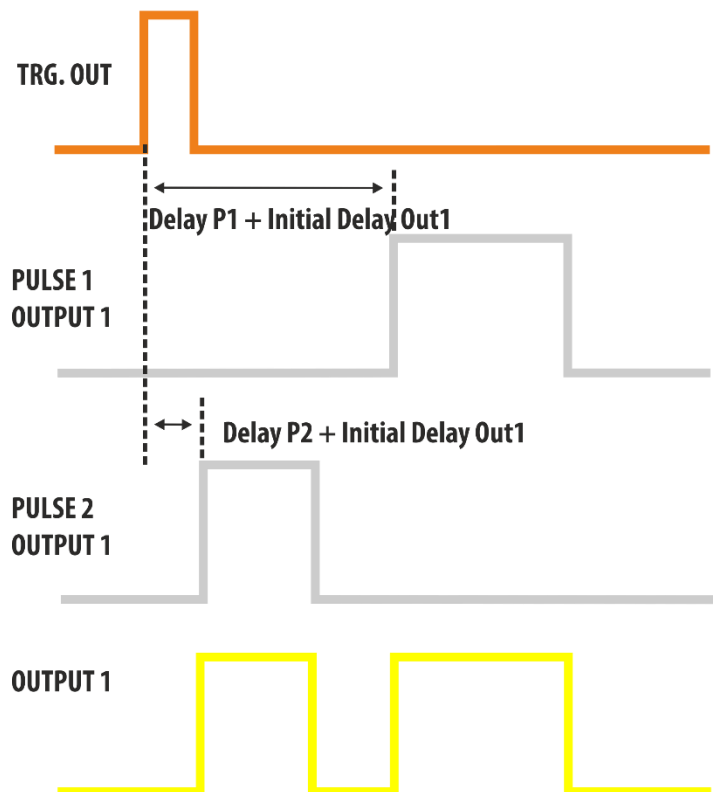
If you press the Settings button on the command bar, you will access to the Output Mode selection. In Continuous and Single modes you will have two, three or four Output pulses for each Trigger Output.

On the right of the page, you can click on the arrow icon to expand or collapse the picture that graphically explains the Output composition in relationship to the Pulses.

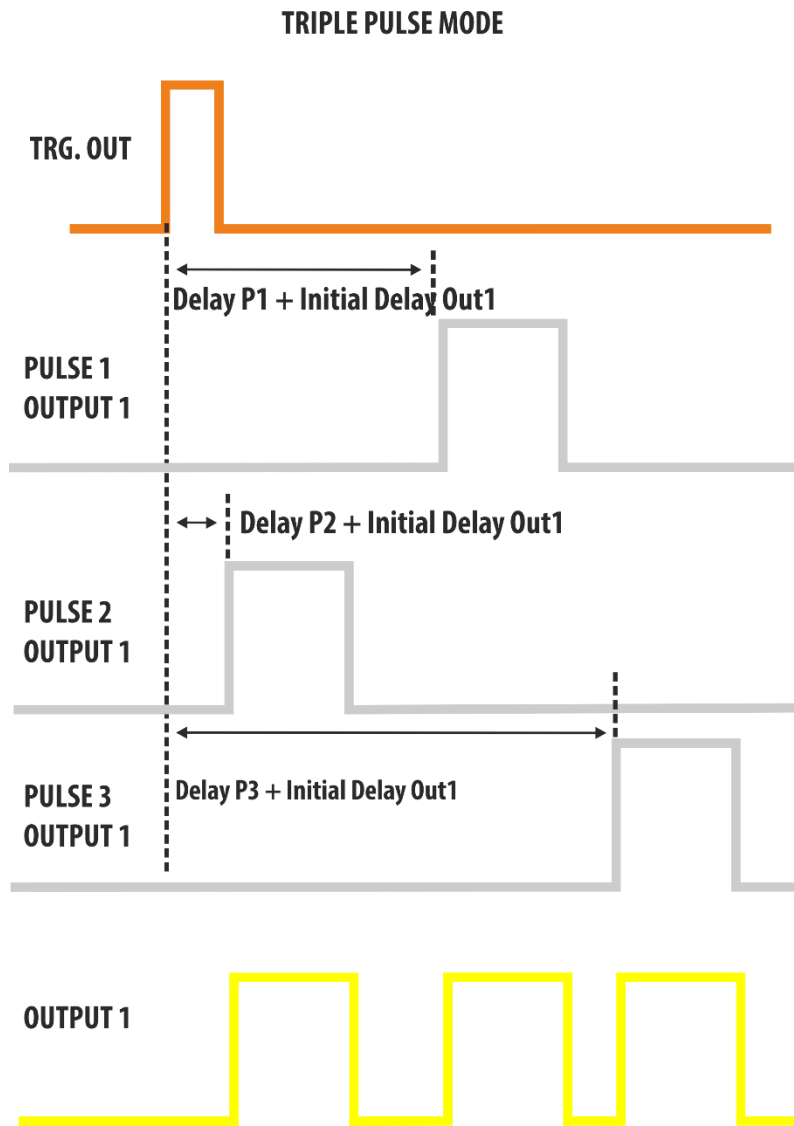


In Double Pulse Mode, the Output signal will be the logical OR combination of the PULSE1 and PULSE2.

DOUBLE PULSE MODE

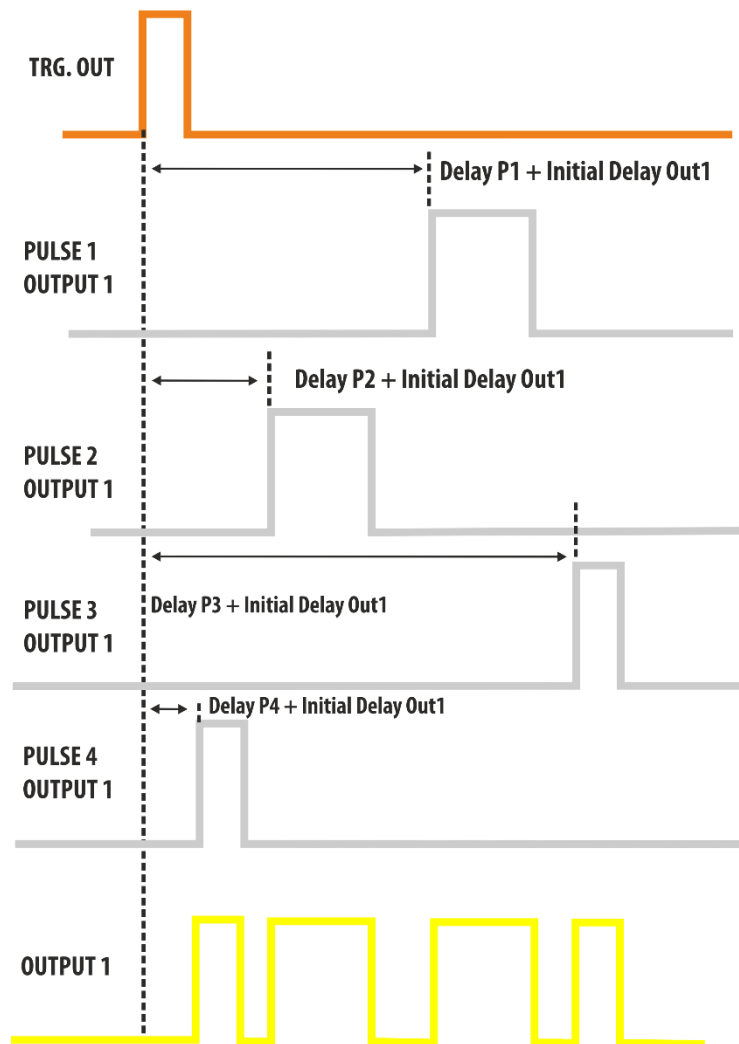


In Triple Pulse Mode, the Output signal will be the logical OR combination of the PULSE1, PULSE2 and PULSE3.



In Quadruple Pulse Mode, the Output signal will be the logical OR combination of the PULSE1, PULSE2, PULSE3 and PULSE4.

QUADRUPLE PULSE MODE



Hint: how to generate a clock

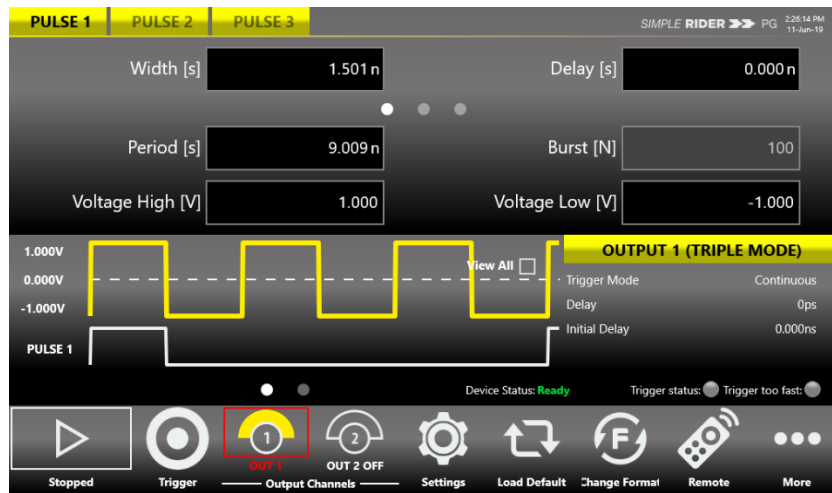
It is possible to create a clock with a frequency greater than 200 MHz using the multiple pulse mode through the following steps:

- Calculate the clock Period. $\text{PeriodClock} = 1/\text{Frequency}$
- Find the lowest value of N that satisfies $N * \text{PeriodClock} > 8 \text{ ns}$ with $N=2,3,4$
- Select Single Pulse mode, set Width = $\text{PeriodClock}/2$, Delay = 0 ns, Period = $N * \text{PeriodClock}$
- Select the Multiple pulse mode that correspond to the N value (2 = Double, 3 = Triple, 4 = Quadruple)

Example: I need to generate a clock frequency of 333 MHz.

1. $\text{PeriodClock} = 1/333 \text{ MHz} = 3.003 \text{ ns}$

2. $N=2 \rightarrow 2*3.003\text{ns} > 8\text{ ns}$? No.
3. $N=3 \rightarrow 3*3.003\text{ns} > 8\text{ ns}$? Yes. The pulse mode will be Triple Pulse Mode.
4. Select Single Mode, Width = $\text{PeriodClock}/2 = 1.501\text{ ns}$, Delay = 0ns, Period = $N*\text{PeriodClock}=9.009\text{ ns}$.
5. Open the Channel Settings and select Triple Pulse Mode
6. Press Run.



Trigger Controls

The Model 765 Outputs are driven from a common timebase and the channel outputs are referenced to the same trigger. It means that the output channels will share the same trigger modes but they can have different period, width and delay.

Trigger Output

The signal available at the Rider Pulse TRG.OUT connector is selectable positive-going or negative-going pulse, synchronized with the TRG.IN signal and the Model 765 internal timebase in Single, Gated and Burst mode.

In Continuous mode the TRG.OUT signal is synchronized with the selected output and the internal timebase.

Trigger Output Delay parameter is relative to the signal's leading edge, the width of the Trigger Output is dependent upon the selected Trigger Mode, its amplitude is programmable from 0.9V to 1.65V into 50 Ohm load.

Trigger Input

The Model 765 can be adjusted to trigger on any signal connected to the TRG.IN connector whose amplitude is greater than 20 mV.

Trigger pulses as narrow as 1 ns can be detected.

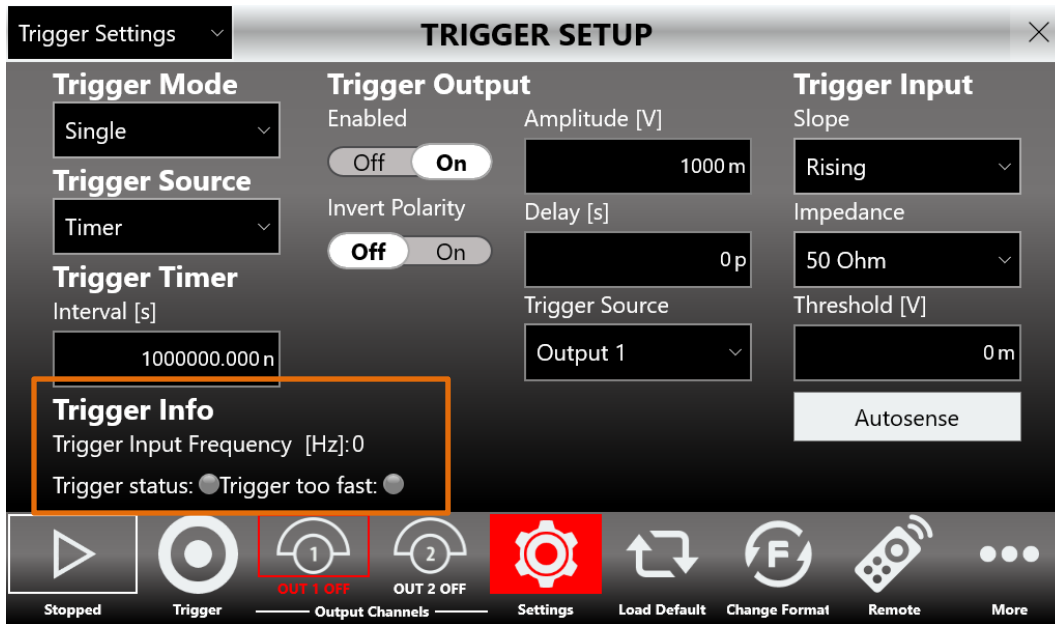
The impedance of the Trigger Input can be programmed to either 1 k Ohm or 50 Ohm.

The signal levels at the TRG.IN must not exceed $\pm 10\text{V}$ into 1 k Ohm and $\pm 3.5\text{V}$ into 50 Ohm; the threshold level can be programmed in the $\pm 8\text{V}$ voltage range with 4 mV steps.

Pressing the button "Autosense", the instrument can automatically measure the threshold of the signal applied to the Trigger In port.

In the externally triggered modes (Singles, Burst and Gated), the Model 765 Frequency Counter measures the frequency of the signal at the TRG. IN and displays the result of the measurement in the Trigger Setup window.

In Single, Burst and Gated mode the software trigger (trigger button) is always active independently from the selected trigger source.



Trigger Modes

The Model 765 has four trigger operating characteristics: Single, Burst, Gated and Continuous.

Single Mode

In Single Mode the Trigger IN source can be sent externally using the TRG.IN connector or internally using the front panel button, SCPI command and timer.

The Trigger Output signal is linked to one of the available Outputs using the **Trigger Output** → **Trigger Source** dropdown list.

The Trigger IN signals starts the Model 765 timebase and one TRG.OUT signal follows each trigger that is detected on the selected output channel with a delay below 100 ns.

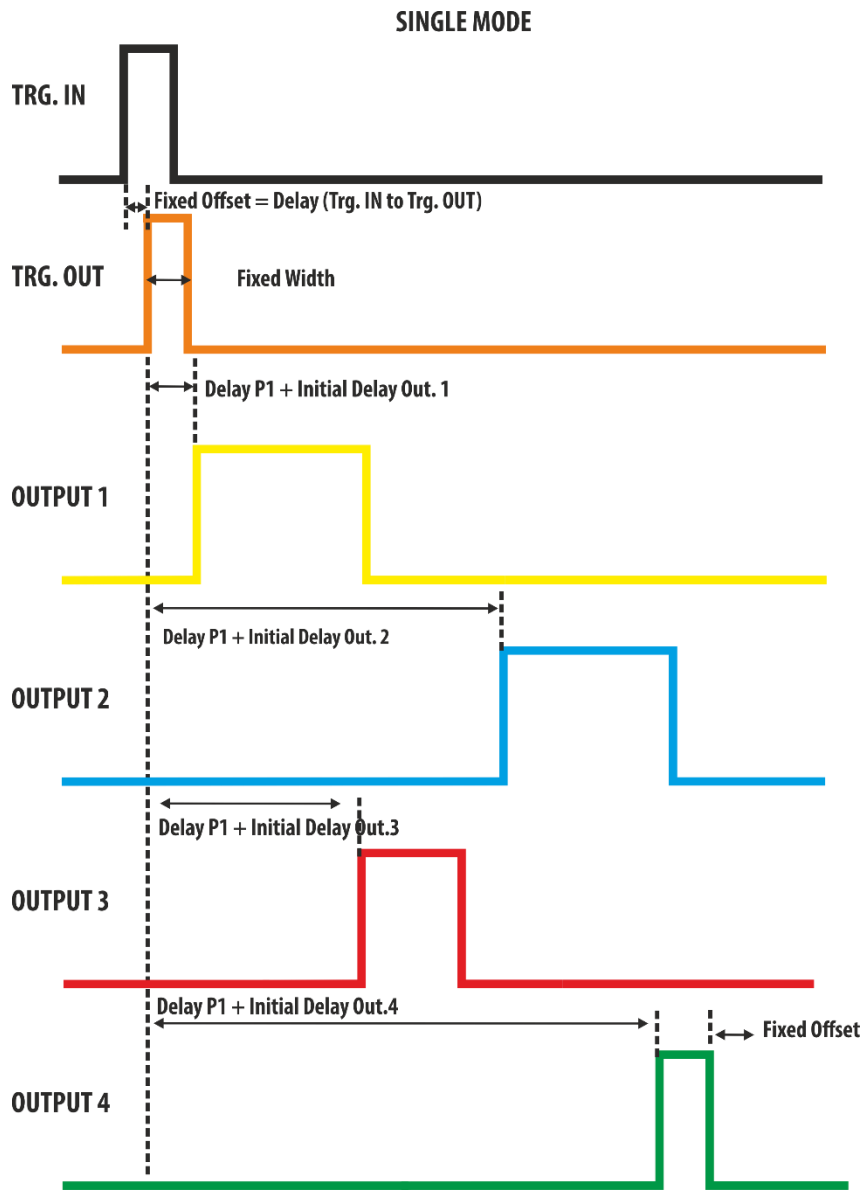
IMPORTANT NOTES

- If the Output channel linked to the Trigger Output Source is disabled (OFF state), the TRG.OUT signal will not follow that output. Its state will be '0' or '1' depending on the selected polarity.
- If all channels will be disabled, the TRG.OUT signal state will be '0' or '1' depending on the selected polarity.

The Output pulses will follow the Trigger Output signals by a Delay user parameter plus an initial channel delay. In this mode the output signals will be generated once.

The Period parameter has no meaning in Single Trigger Mode and the output pulse duration will be determined by the Delay plus the Width parameters.

The Trigger Output width is fixed and its duration is about 10 ns.



Burst Mode

Burst mode is similar to Single mode, but a programmed number of Output Pulses is generated for each trigger signal that it is detected on the selected output, rather than just once.

The Trigger Output signal is linked to one of the available Outputs using the **Trigger Output → Trigger Source** dropdown list.

This number can be programmed from 1 to 4294967295 by selecting the Burst[N] parameter on the user interface.

The trigger source can be sent externally using the TRG.IN connector or internally using the front panel button, SCPI command and timer.

The Trigger IN signals starts the Model 765 timebase and one TRG.OUT signal follows each trigger that is detected on the selected output channel with a delay below 100ns.

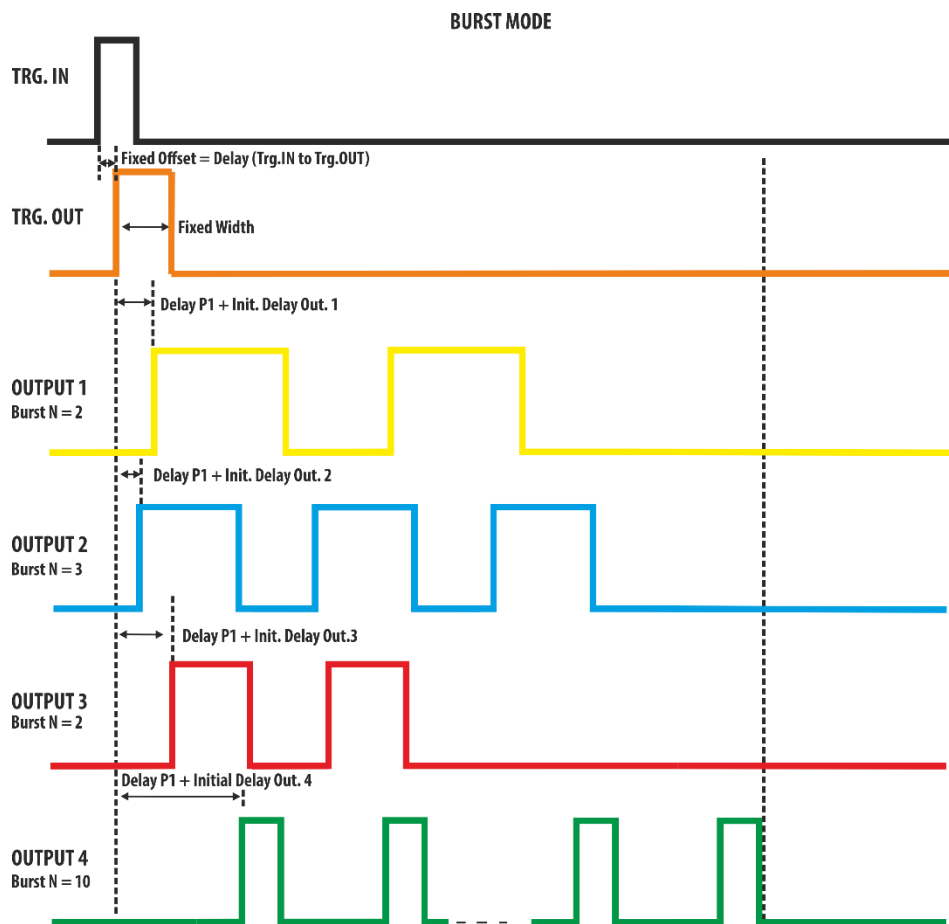
IMPORTANT NOTES

- If the Output channel linked to the Trigger Output Source is disabled (OFF state), the TRG.OUT signal will not follow that output. Its state will be '0' or '1' depending on the selected polarity.
- If all channels will be disabled, the TRG.OUT signal state will be '0' or '1' depending on the selected polarity.

The Output pulses will follow the Trigger Output signals by a Delay user parameter plus an Initial channel delay parameter. In this mode the output signals will be repeated by the number of times specified in the Burst parameter.

The time between Output pulse is specified by the Period and the duration of each Output pulse is specified by Width.

The Trigger Output width is fixed and its duration is about 10 ns.



Gated Mode

In Gated mode, the Output Pulses of the programmed Width occur at the rate specified by Period as long as the Trigger signal is true.

The trigger source can be sent externally using the TRG.IN connector or internally using the front panel button, SCPI command and timer.

The Trigger signals starts the Model 765 timebase and one TRG.OUT signal follows each trigger by a Fixed Offset equal to the Trigger IN to Trigger OUT Delay.

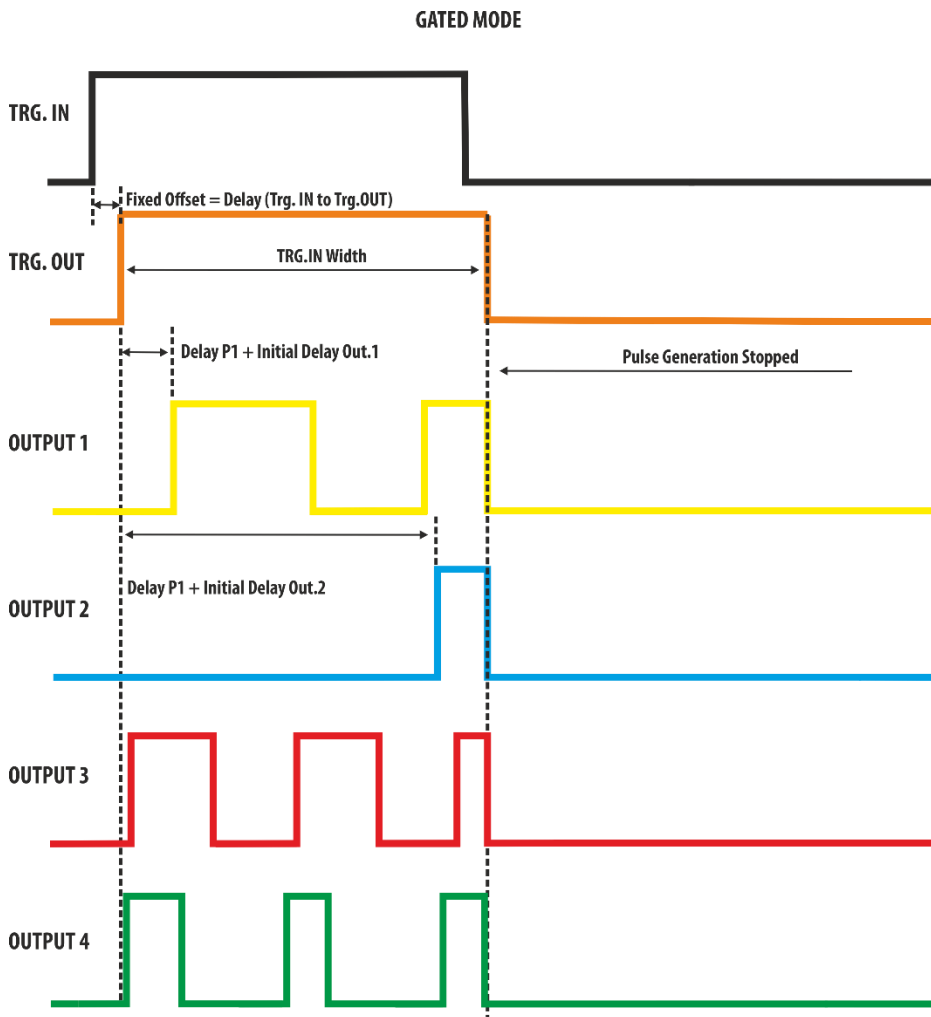
IMPORTANT NOTES

- If the Output channel linked to the Trigger Output Source is disabled (OFF state), the TRG.OUT signal will not follow that output. Its state will be '0' or '1' depending on the selected polarity.
- If all channels will be disabled, the TRG.OUT signal state will be '0' or '1' depending on the selected polarity.

The width of the Trigger Output will be equal to the width of the Trigger IN signal.

In Gated mode the Output pulses start after the Delay parameter plus a fixed offset below 100 ns and continue to run at the rate defined by Period, for the duration of the true state of the Trigger signal.

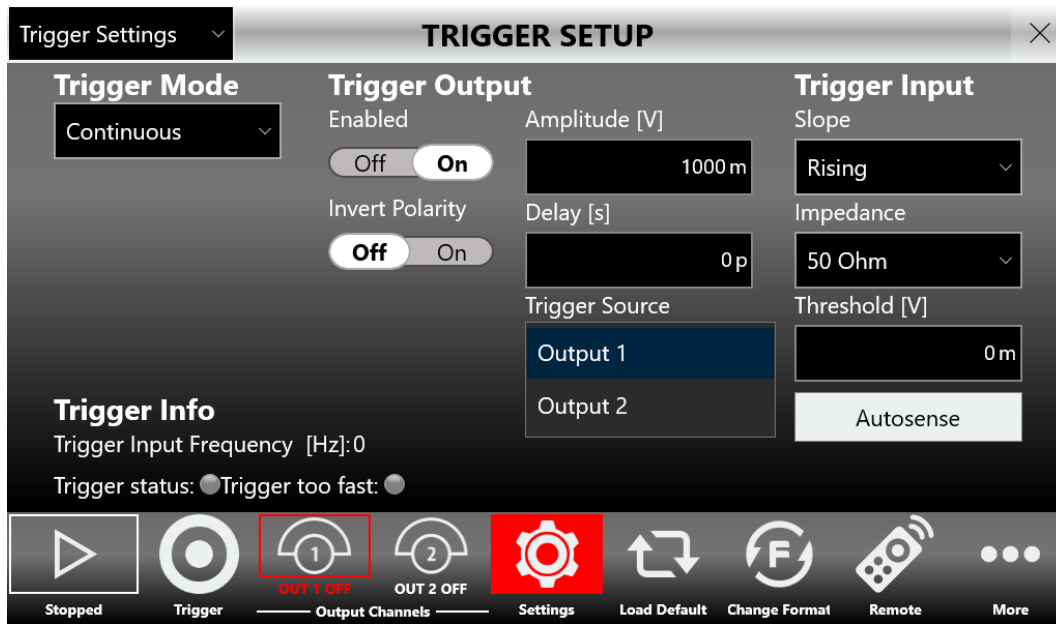
If the Trigger Output signal goes false while an Output pulse is active, the generation of the output pulses will be stopped.



Continuous Mode

Continuous mode produces a continuous pulse stream at the selected Period and Width.

One Trigger Output pulse will occur for each Pulse 1 relative to the selected Output channel in the Trigger Source dropdown list.

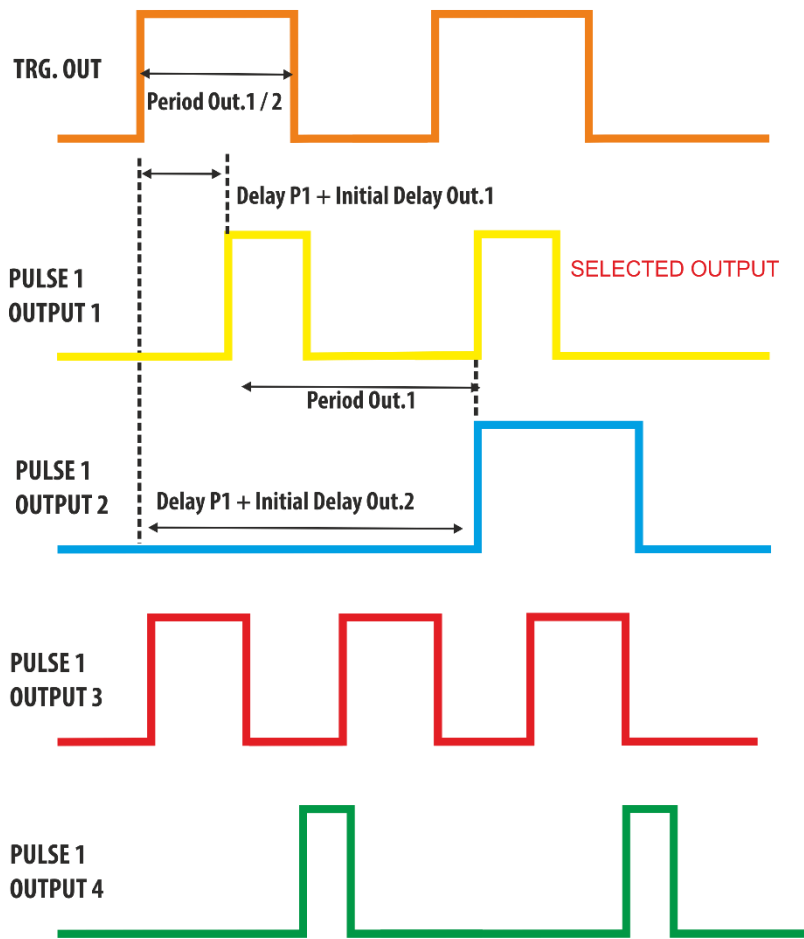


IMPORTANT NOTE: if the selected output channel for Continuous mode will be disabled (OFF state), the TRG.OUT signal state will be '0' or '1' depending on the selected polarity.

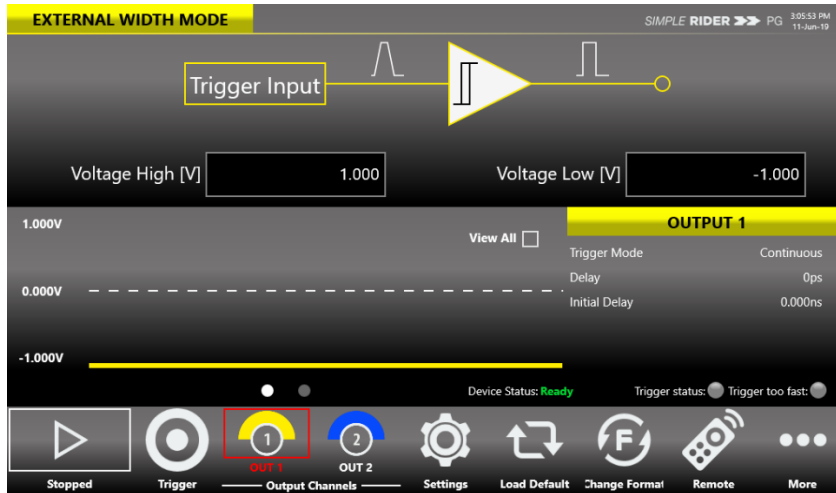
The width of the Trigger Output will be equal to the Period of the selected output divided by 2.
 Trigger Output Width = Period / 2 of the selected output.

The Output Pulse follows the Trigger Output by a Delay parameter. In the picture below the Trigger Out signal is linked to the Pulse 1 of the Output 1.

CONTINUOUS MODE

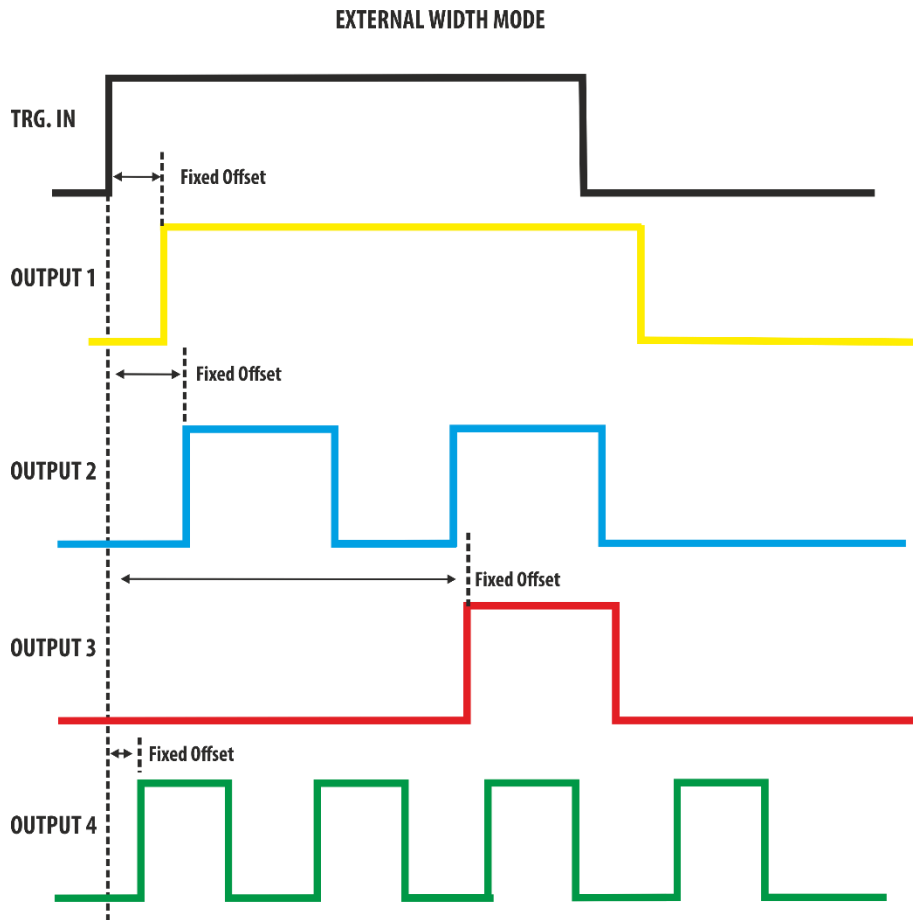


External Width Mode



In External Width mode, the TRG.IN signal (SMA connector) is redirected to the Output channel with programmable voltage level, threshold and output impedance.

NOTE: the TRG.OUT signal in Single, Burst and Gated mode will follow the output channels where the External Width Mode is not selected; in Continuous mode if the selected source is External Width, the Trigger Out frequency will be the same of the Trigger In signal.



Multiple Pulse Interactions

Here below is described the interaction of multiple pulses with various Trigger Modes.

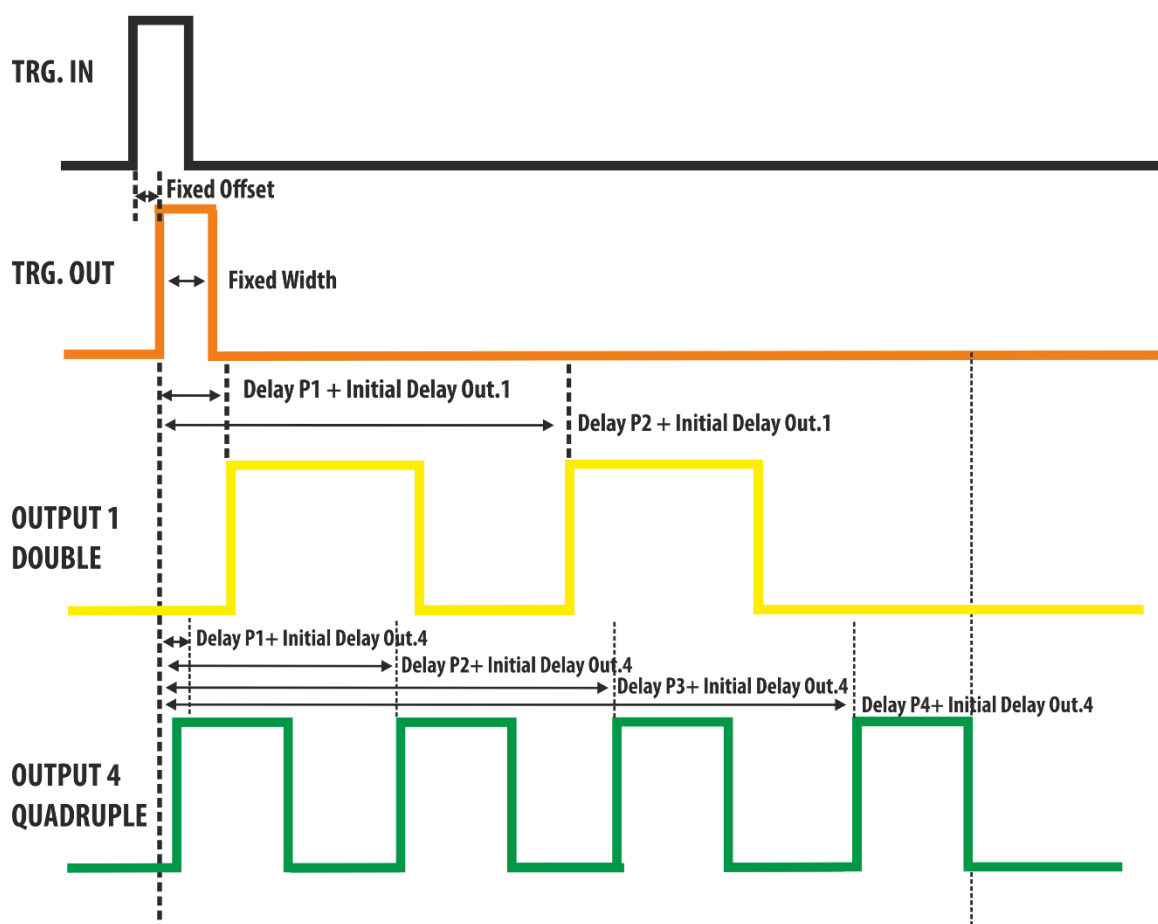
Continuous Mode

Depending on the multiple pulse selection, two, three or four Pulses follow each Trigger Output rather than one.

Single Mode

Same as Continuous mode for the exception that the group of pulses are generated only once.

SINGLE MODE: MULTIPLE PULSE



Burst Mode

In this case the Burst N parameter specifies the number of Output Pulse groups (two, three or four) per trigger, rather than the number of Output Pulses.

As example in case of Double Pulse, the first pulse of the first pair in any burst will follow the leading edge of the trigger Output **by Delay 1 + Initial Delay Out.1**, the second pulse in any pair follows the first by Delay parameter, and the first pulse of any pair follows the first pulse of the previous pair by the Period parameter.

Gated Mode

Same as Continuous mode for the exception that the group of pulses are generated when the Trigger signal is in true state.

External Width Mode

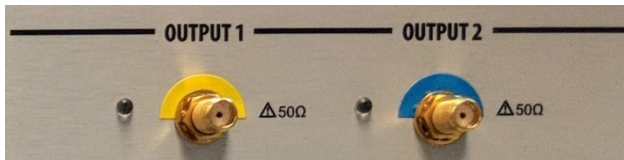
Its function is the same of single pulse, it is unaffected by the state of Multiple Pulses.

Quick Start

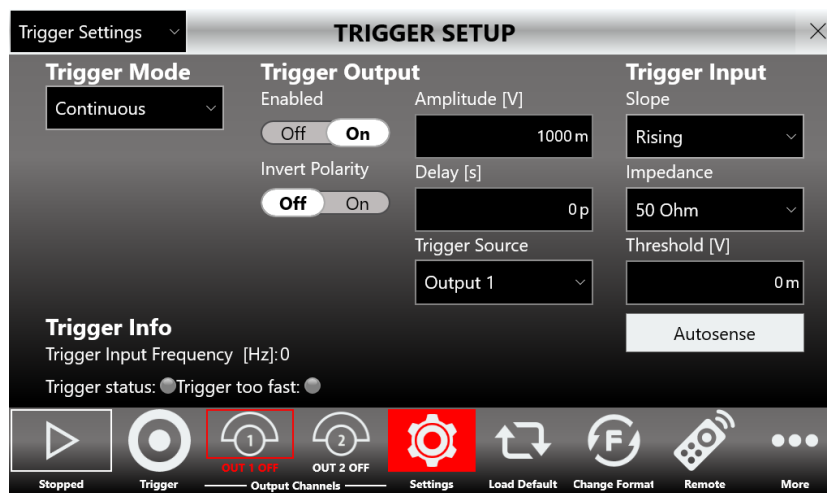
Single Pulse

If you are a beginner user, you can follow the steps here below to generate your first pulse signal.

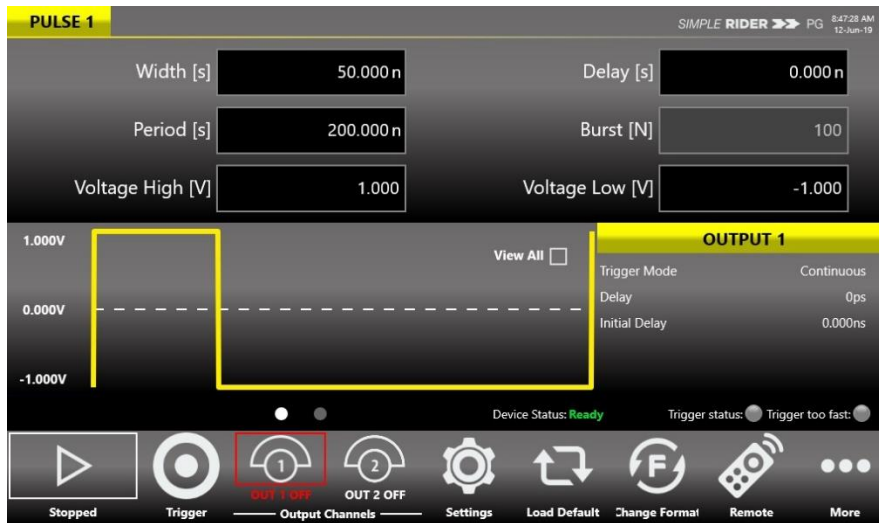
1. Connect the power cord and the push the front-panel on/off switch  to turn on the instrument.
2. Connect the Output 1 of the instrument to the oscilloscope input with a cable, select 50 Ohm load on the oscilloscope input.



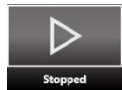
3. Touch the  button on the SimpleRider UI.
4. In the Settings → Trigger Setup page, select Continuous as Trigger Mode.



5. By default the OUT1, OUT2 are in OFF state: it means that the output is disconnected from the load using a relay.



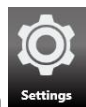
Press the OUT1 button to enable it, press OUT2 to select OUTPUT2 channel, press it again to enable it.



6. Press the Start button
7. Observe the pulse waveform displayed on the oscilloscope screen.
8. Press again the Settings button to close the Settings page.
9. Touch the Voltage High [V] to set the Output pulse high level.
You can also select the Period, Delay or Width parameter using the display touch screen or the front panel physical buttons.
10. Change the Output pulse parameters using the virtual numeric keypad or the physical numeric keypad, the general purpose knob and the arrow keys, or, keyboard and mouse.

Double Pulse

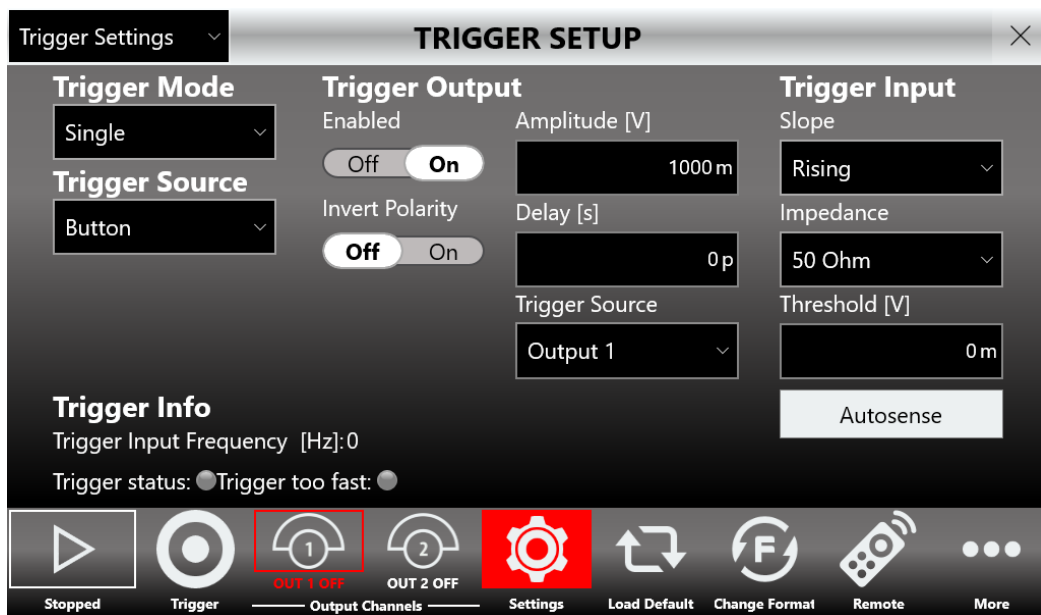
1. Connect the power cord and the push the front-panel on/off switch  to turn on the instrument.
2. Connect the Output 1 of the instrument to the oscilloscope input with a cable, select 50 Ohm load on the oscilloscope input.



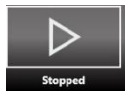
3. Press the Settings button, select "Outputs Settings" in the dropdown menu and select Double Pulse option in the Pulse Mode relative to the Output 1



4. Select "Trigger Settings" in the dropdown menu to open the Trigger Setup page and select *Single* as Trigger Mode, *Button Only* as Trigger Source.



5. Press the Start button

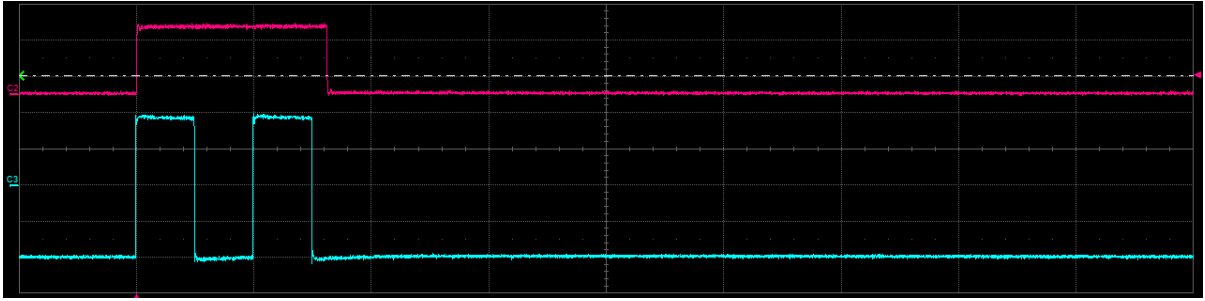


6. Set a Single acquisition on the oscilloscope, set the scope trigger and press the Trigger



button . Each time you will press the Trigger button you will observe a group of two

pulses displayed on the scope.

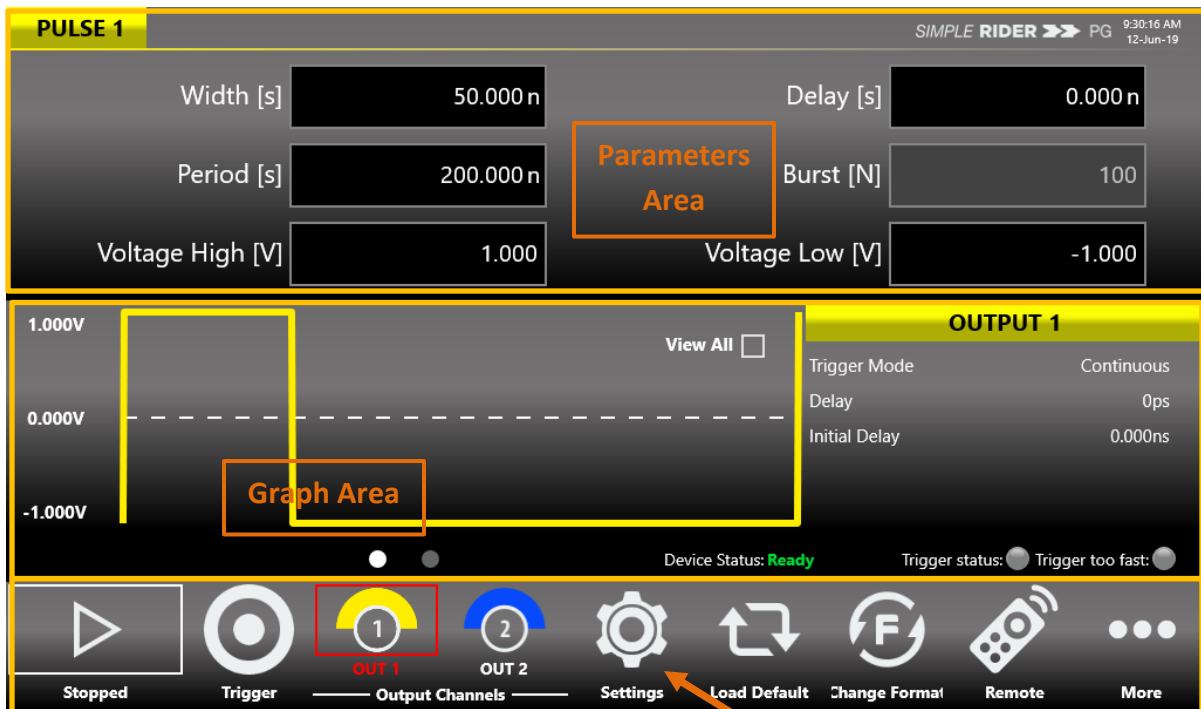


7. Press again the Settings button to close the Settings page.
8. Touch the Output 1 → Pulse 1 Delay parameter to change the Pulse 1 delay.
You can also select the Amplitude, Period or Width parameter using the display touch screen or the front panel physical buttons.
9. You can use the swipe gesture on the top of the display to change the Pulse selection from Pulse 1 to Pulse 2 and viceversa.
10. Change the Output and the Pulse parameters using the virtual numeric keypad or the physical numeric keypad, the general purpose knob and the arrow keys, or, keyboard and mouse.

Simple Rider PG Software

User Interface Description

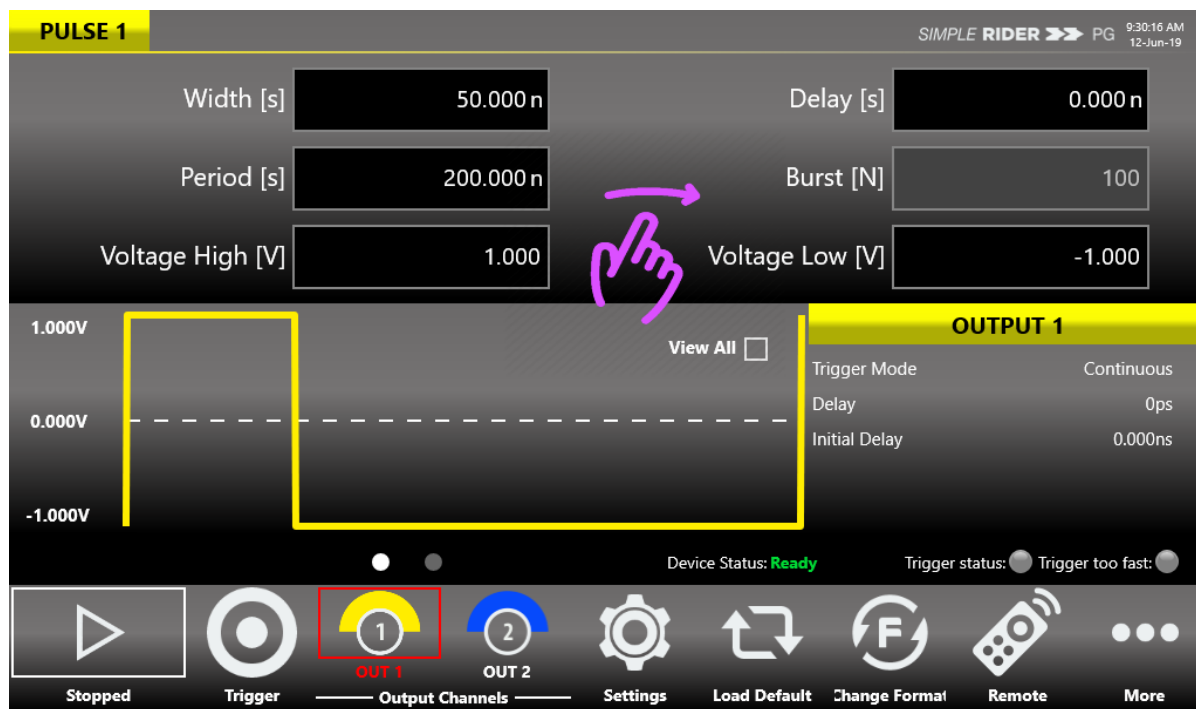
The SimpleRider software environment provides an easy access to all product functions and parameters.



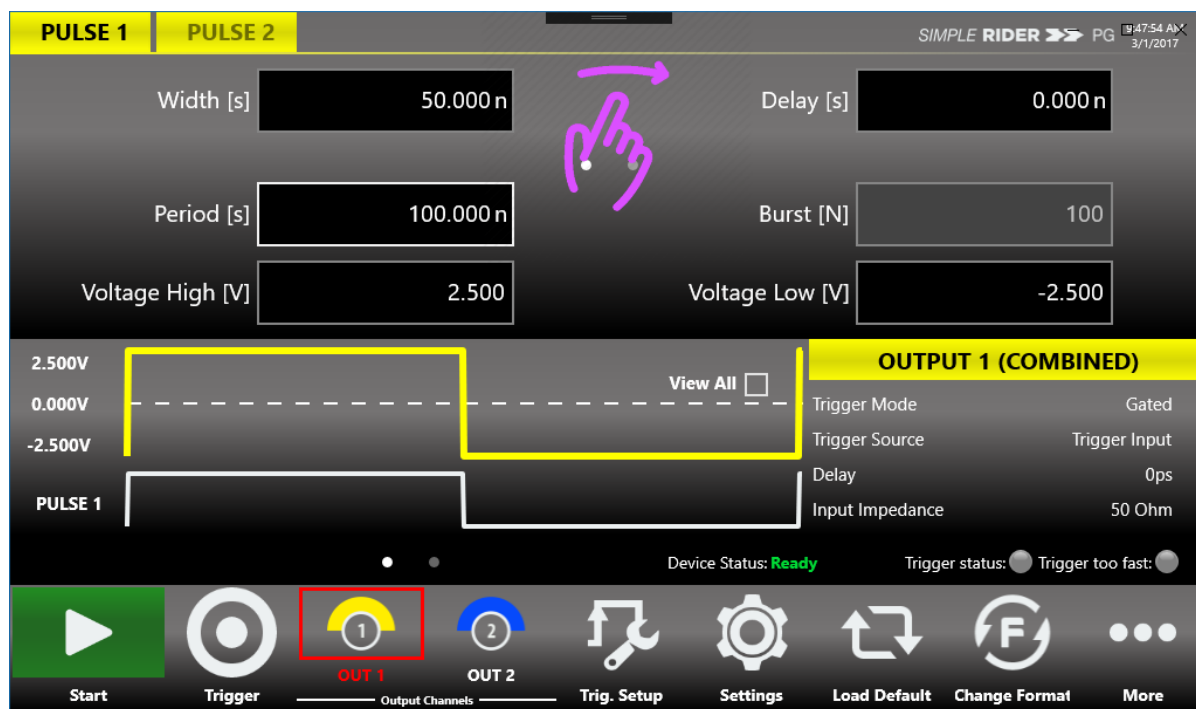
It consists of three main elements:

- The Pulse and Output channel parameters area
- The Waveform Graph area
- The command bar

As mentioned, the display is 7" capacitive touch screen display and you can have access to parameters and generate pulses quickly by a few screen touches.



If Multiple Pulse feature is Off, you can use the Swipe left or right gesture to change the Output channel selection.



If the Multiple Pulses feature is On, you can swipe left or right on the upper part of the screen to change the selection of the Pulse and swipe left or right in the lower part of the screen to change the Output selection.

Parameters Area

If Multiple Pulses features is off, the parameter area gives you access to the Pulse 1 (Width, Delay) and to the selected Output parameters in terms of Period, Voltage High, Voltage Low and Burst number.

PULSE 1		SIMPLE RIDER >>> PG 9:39:12 AM 12/14/2016	
Period [s]	200.000 n	Width [s]	100.000 n
Initial Delay [s]	0 p	Burst [N]	100
Voltage High [V]	2.000	Voltage Low [V]	-2.000

You can touch the parameter area to open the Virtual numeric keypad, edit the parameter value and its measure unit.

The screenshot shows the virtual numeric keypad interface. Callout 1 points to the parameter name, value, and unit display area (e.g., 'Voltage High [V] 1.200'). Callout 2 points to the numeric keypad grid. Callout 3 points to the unit selection buttons (mV, V). Callout 4 points to the 'Close', 'Bksp', 'Delete', and 'Enter' buttons. Callout 5 points to the 'COARSE' and 'Fine' toggle buttons. Callout 6 points to the horizontal slider at the bottom.

1. This area of the virtual keyboard displays the parameter name, value and unit measure. You select the parameter digit you want to modify by touching the digit position on the display or the left/right arrow in the area numbered as 3; the selected digit will be highlighted in blue. You can also use the front panel $\leftarrow$ and $\rightarrow$ arrows.
2. Each number pressed will be displayed in the area 1, the [+/-] key will toggle the sign of the number being entered and may be pressed at any time before terminating entry.

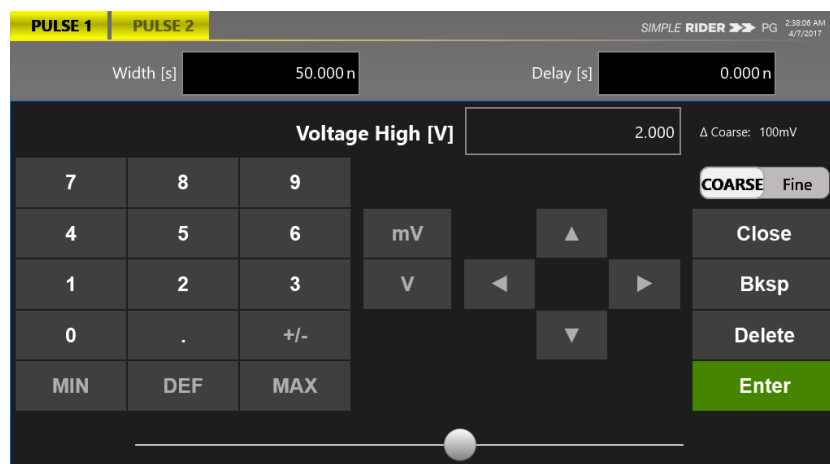
Touch the MIN. and MAX. buttons to display the minimum and maximum allowed value for the selected parameter; use the DEF. button to set the default value.

3. After the sign and numeric portion of the desired value have been punched in, entry with the appropriate multiplier is terminated by pressing one of the unit measure keys that are available for the current parameter.

You can select a parameter digit using left/right arrows or increment/decrement the value using up/down arrow.

4. Coarse / Fine: the coarse/fine button let you to modify the granularity of the increment. When Fine is pressed, the increment is set to 1 for the selected digit. You can increment or decrement the selected parameter using the UP/DOWN arrows button or rotating knob on the front panel.

When Coarse is pressed, the Delta increment is displayed in the parameter area and the parameter value changes in steps of the selected increment without taking account of the highlighted digit.



You can keep pressed the rotating knob and rotate it on the right or on the left to change the Delta increment.

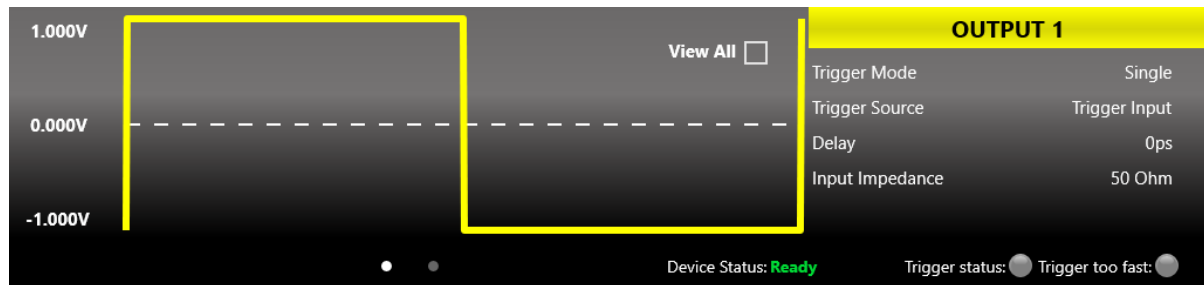
5. The Close button closes the virtual keypad without applying any changes on the instrument while the Enter button confirms the changes and it applies them on the instrument. Bksp button is provided for deleting erroneous key presses, Delete button deletes all the entry.
6. The horizontal scrollbar allows to change quickly the selected value. The position specify the value between the minimum and the maximum.
The increment/decrement value entered by the rotary knob or by the scrollbar are applied on the instrument on the fly.

Notes

- In Single Mode the Period and Burst parameters are not available.
- In Gated and Continuous modes, the Burst parameter is not available.

Graph Area

The graph area displays the Output channel and/or the Pulse waveforms with a vertical legend containing the voltage high, low level and offset.



The right side of the graph displays the channel name and a list with all the main current instrument settings like Trigger Mode, Source and Amplitude, Trigger Input Impedance etc.

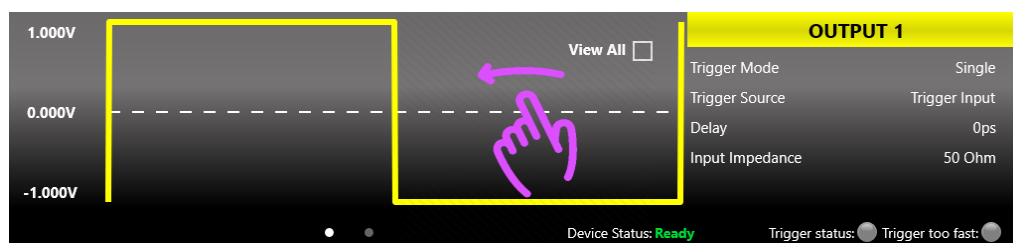
In the bottom part of the Graph Area, there are the *Device Status*, *Trigger Status* and *Trigger Too Fast indicators*.

If the **Device Status** is "Ready", it means that the instrument is stopped and ready to start, when you press the Run button the Device Status will change to "Running" and it means that the device is ready to receive a Trigger signal.

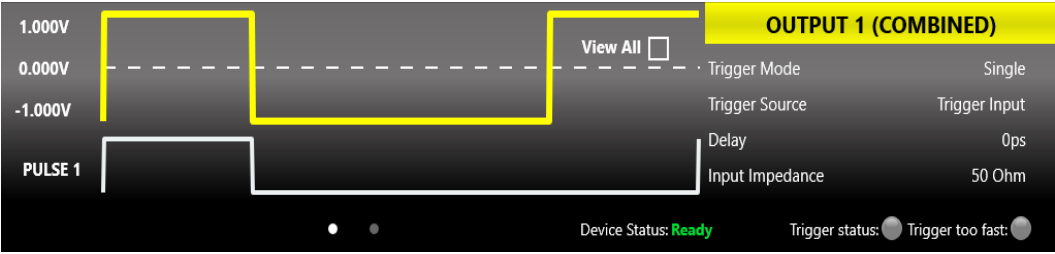
The **Trigger status** led notifies you when the instrument received the trigger signal.

The **Trigger too fast** led notifies you that the trigger event has been latched, but the trigger frequency is too high and the instrument cannot be rearmed before the completion of the previous trigger event, so in this case some trigger events may be lost.

Note that you can use the swipe gesture in this area to change the Output channel selection.



If the Double, Triple or Quadruple output pulse mode option is selected, the Graph Area will display in the upper area the Output signal graph and in the lower area the selected pulse graph.



Command Bar Area

The command bar contains several touch buttons to control the instrument, the following table describes them in detail.

	START/STOP button - Use this button to set the instrument in Running state or Ready (Stopped state). • If the instrument is in <i>Continuous trigger mode</i>, when you press Start the instrument will start the Output pulses generation. If you press again the button, it will stop the generation. • In <i>Single, Gated and Burst mode</i> when you press this button the instrument will be armed to receive the Trigger signal. As soon as the Trigger signal will be received, the Output pulses will be generated. If the instrument is in running state and you press the Start/Stop button, the output pulse generation will be stopped and the instrument will not be no more armed to receive the Trigger.

	Trigger Button – Use this button to send an internal software trigger to the instrument. If the instrument is armed and the trigger source is Internal, as soon as the Model 765 receive the software trigger, the output pulses will be generated.
	Output Channels Buttons - Press OUT1, OUT2, OUT3 and OUT4 button (1074 models) to change the Output channel page. If you press again on the selected Output,  you can turn OFF/ON : the output will be disconnected by using a relay.
	Settings Button – Use this button to open the Settings Page. From the Settings Page you can have access to the Trigger, Output and Clock settings.
	Load Default – Use this button to load the instrument default settings (two-channels version only).
	Change Format – Use this button to change the format of the output and pulse parameter (two-channels version only). Format A is: Period, Width, Delay, Phase, Voltage Level High, Voltage Level Low. Format B is: Frequency, Duty Cycle, Phase, Amplitude, Offset.
	Remote Button – Use this button to entry or exit from Remote Mode (two-channels version only).
	More Button – Use this button to have access to other instrument features.
More Button Menu Items	
Exit 	Exit Button – Press this button to close the application.
Exit Full Screen 	Exit Full Screen Button – press this button to exit from the full screen mode; in this way you can access to Windows OS.

Change Format 	Change Format – Use this button to change the format of the output and pulse parameter (four-channel version only).
Remote 	Remote Button – Use this button to entry or exit from Remote Mode(four-channel version only).
Calibration 	Calibration Button – Use this button to perform the calibration and diagnostic.
Default 	Default Button – Use this button to load the default instrument settings (four-channel version only).
Load From 	Load From Button – Use this button to load a configuration file. (For more information, please refer to the relative section).
Save As 	Save As – Use this button to Save the Current configuration into an existing one or create a new one. (For more information, please refer to the relative section).
Export 	Export – Use this button to export the current configuration. (For more information, please refer to the relative section).
Keyboard 	Numeric Keyboard Button – Use this button to enable or disable the virtual numeric keyboard.
Beep 	Beep Button – Use this button to enable or disable the beep audio signal when the user touches a button.
About 	About Button – Use this button to check the credits, the software, the firmware release number and the instrument serial number.
Help 	Help Button – Use this button to open the User Manual

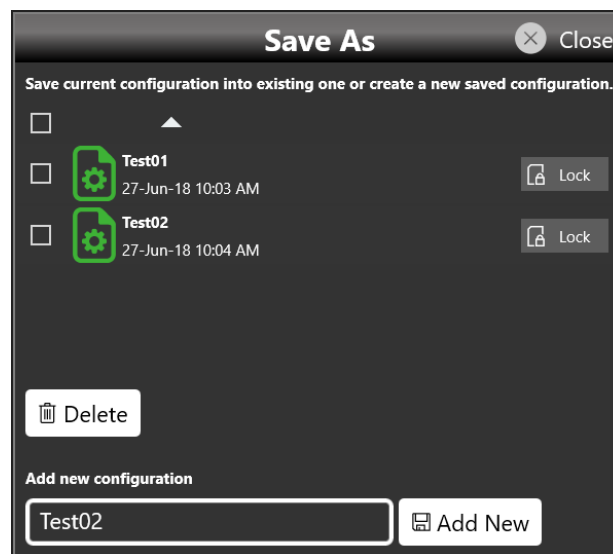
Configurations

A configuration contains the data in proprietary format relative to the current instrument settings. You can save and load the instrument settings by using the “Save As..” and “Load From...” buttons.

Save As...



A configuration can be saved by means of the “Save As” button that will open the following dialog box. The configuration will be saved in the configuration list that can be accessed by the “Load From” dialog box



In this page you can add a new configuration entry or overwrite an existing one. To create a new configuration entry, it is necessary to write a name in the text box in the bottom of the page and click on “Add New”.

Export Configuration



If you touch the Export Configuration button a proprietary binary .zip file relative to the current configuration will be exported. The exported file can be used to share configurations between different users or instruments.

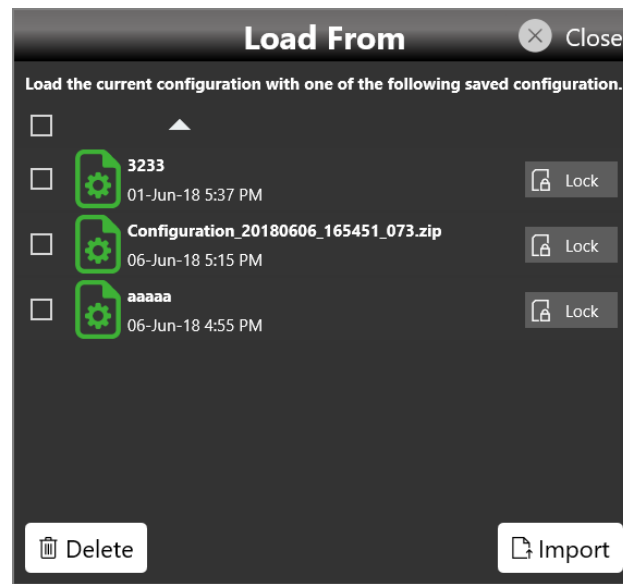
Load From...



Touching the “Load From” button in the “More” menu, a page will be opened that shows the list of all the saved and imported configurations.

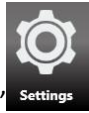
If you select an existing Configuration, you will load all the settings into the instrument.

In the “Load From” page it is also possible to manage the configuration list: you can delete, import or lock a configuration. When a configuration is locked it can’t be deleted or overwritten.



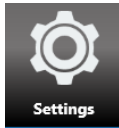
If you touch the Import Configuration button you can import the configuration file that comes from a different machine or from a different user. The imported configuration will be inserted in the “Load From” list.

Settings



The “Settings button” allows you to have access to the Output, Trigger and Clock Settings pages.

Trigger Setup



Press this button and select “Trigger Settings” on the dropdown list to open the Trigger Setup page where you can select the *Trigger Mode*, the *Trigger Source*, the *Trigger Output Amplitude and Delay*, the *Trigger Input* and the *Timer* characteristics.

Trigger Control	Description
Trigger Mode	The Model 765 has four trigger operating characteristics: Single, Burst, Gated and Continuous. Please read the Trigger Modes section for a detailed explanation.
Trigger Source	<i>External</i>: the instrument receives the trigger from the TRG.IN SMA connector AND from the Trigger button. <i>Timer</i>: the instrument trigger is linked to the software internal Timer AND to the Trigger button. You can set the timer value from 20 ns to 50 seconds; when you press the Start button the countdown timer will start and as soon as it will reach the zero value, a trigger signal will be generated.

Trigger Input	In Single, Burst and Gated mode, you can set the Trigger Input Threshold, the slope (positive or negative) and the input impedance (50 Ohm or 1 k Ohm). Pressing the button “Autosense”, the instrument can automatically measure the threshold of the signal applied to the Trigger In port. Trigger Input Slope Rising ▼ Impedance 50 Ohm ▼ Threshold [V] 0 m Autosense
----------------------	---

Output Settings



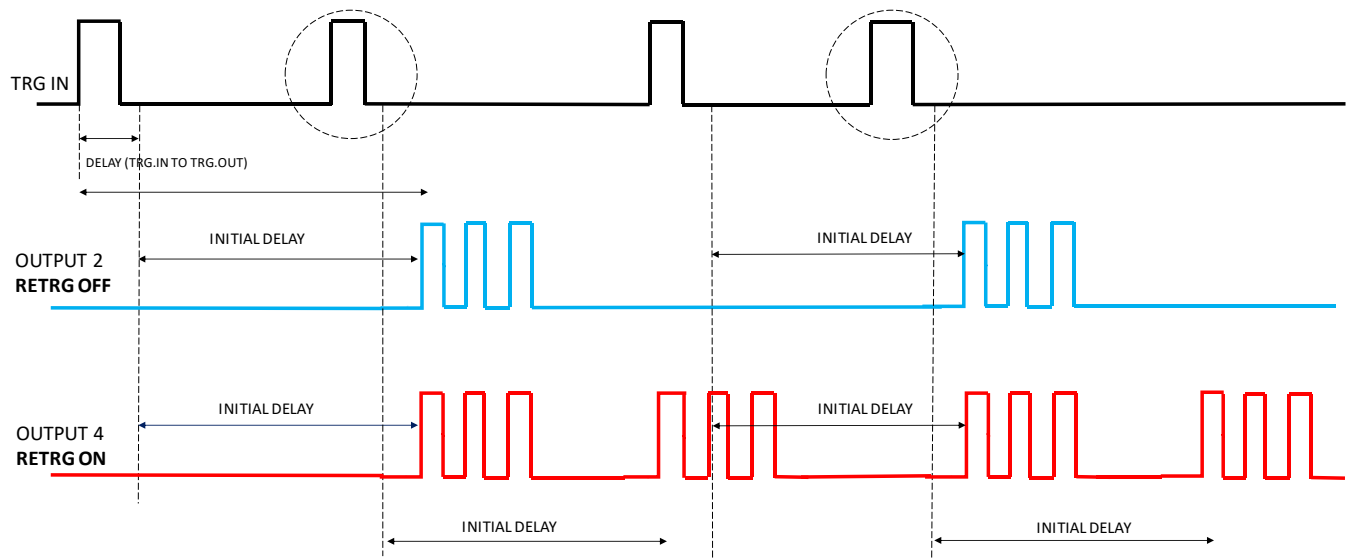
Press this button , select “Outputs Settings” on the dropdown menu to open the Settings page where you can set the output channel features like selecting the pulse mode (single, double, triple or quadruple), setting the load impedance, the output polarity.

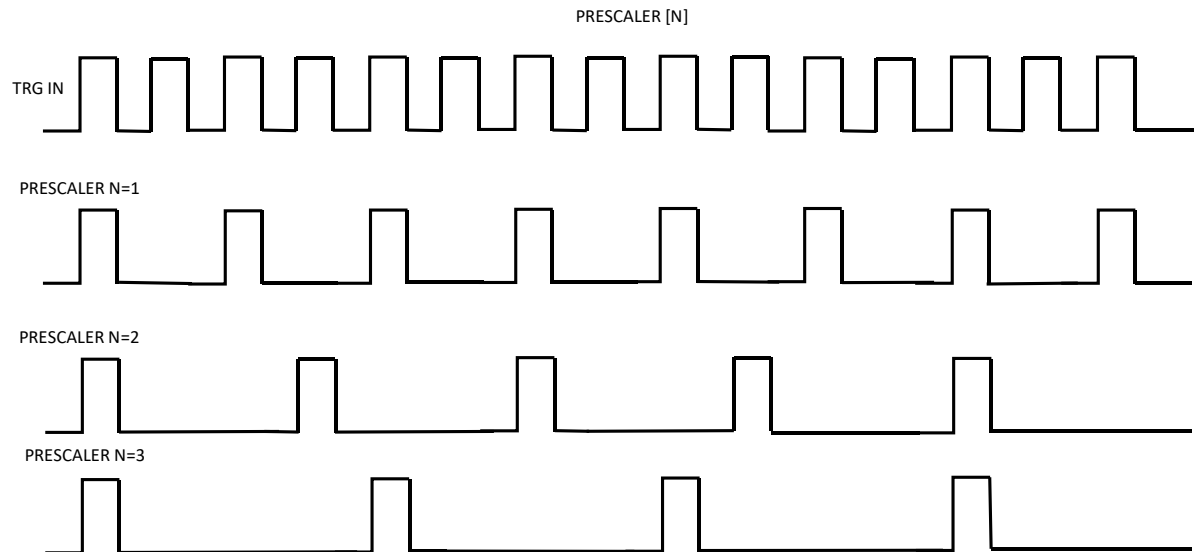


Control	Description
Pulse Mode	It selects the pulse mode, the pulse mode can be single, double, triple and quadruple. If you need to obtain the maximum repetition rate (800 MHz), you should select the Quadruple mode, period 5 ns and select a width of 0.625 ns on each pulse, set Delay 0ns on Pulse 1, Delay 1.25 ns on Pulse2, Delay 2.5ns on Pulse 3 and Delay 3.75ns on Pulse 4.
Invert	When “Invert” is ON, it inverts the output polarity.
Impedance Value [Ohm]	If the “Auto Load Compensation” is Off, you should specify the value of the load impedance attached to the selected Output channel connector.
Auto Load Compensation	When “Auto Load Compensation” is turned On, the generator will make automatically the corrections to deliver the displayed voltage levels to any load connected to the selected Output within the 50 Ohm to 100 M Ohm.
Initial Delay[s]	It is the Initial delay of the selected channel from the Trigger Output signal. The Output pulses will follow the Trigger Output signals by a Delay user parameter plus the Initial Delay parameter.
Retriggerable Delay	If the “Retriggerable Delay” is Off, as soon as it arrives the Trigger event the instrument will wait for the Initial Delay value before starting the output pulse generation. Other trigger events that may occur during the Initial Delay will be ignored.

	If it is “On”, the instrument can record all the trigger events that arrive during the Initial Delay for a maximum of 2.5 μs. In this way the trigger events that occur during the Initial Delay can be detected. This mode allows you to have the maximum trigger rate independently from the Initial Delay that you set.
Trigger Prescaler[N]	This parameter indicates the number of Trigger IN edges that will be discarded by the instrument from the last valid edge. It is a value from 0 to 65535 where ‘0’ means that no edges will be discarded, ‘1’ means that 1 edge will be discarded and so on; as example this can allow you to decrease the Trigger IN rate.

RETRIGGERABLE DELAY

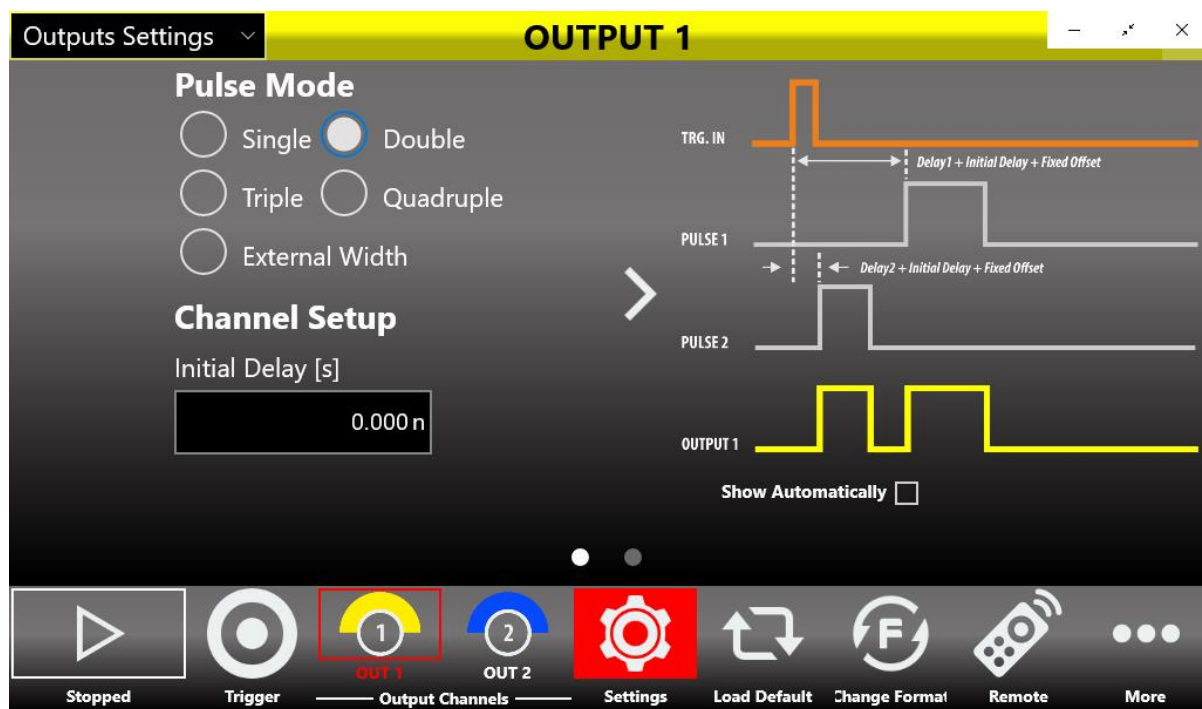




If you press the Left arrow on the right of the screen or you change the output pulse mode, the pulse mode flyout panel will be opened.

This flyout panel explains graphically how the pulses are combined together to generate the output signal and the timing relationship between the Trigger Output, the Pulses and the selected output.

You can press the Right arrow to collapse the view panel.



Clock Settings



Press this button , select “Clock Settings” on the dropdown menu to open the Clock Setup page.



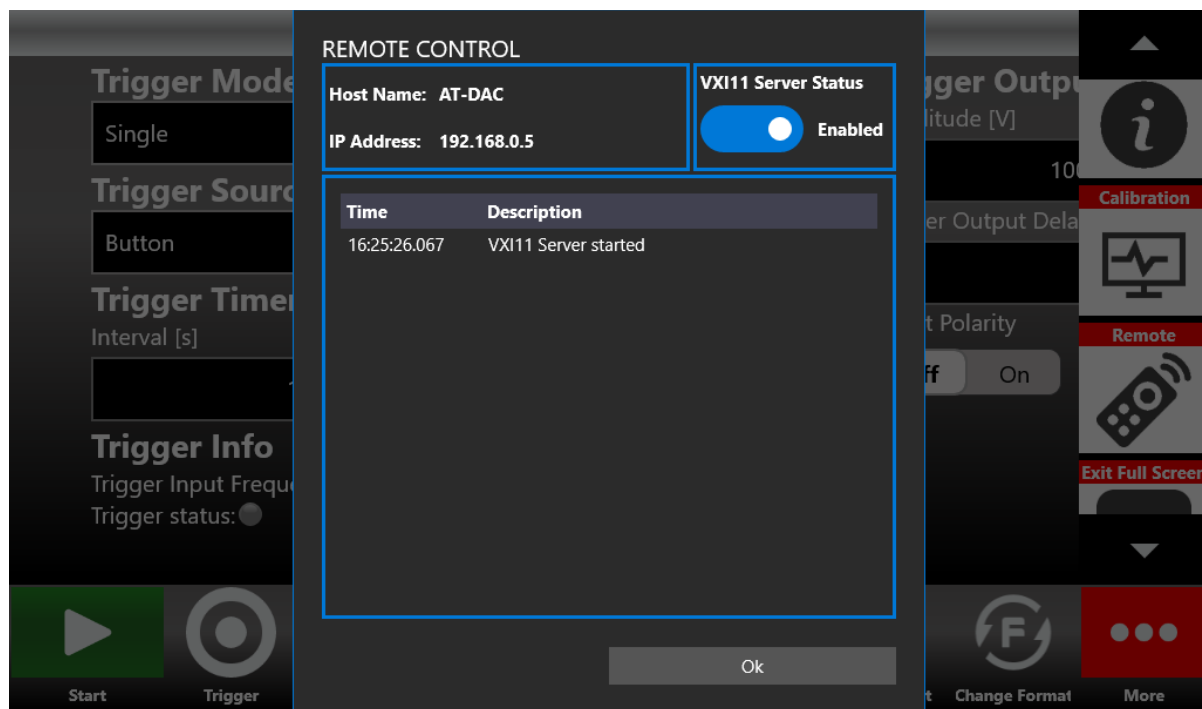
Clock Source: specifies the clock source as Internal or External.

If **Internal Clock** is selected, the instrument time base is synthesized using a reference clock generated internally.

If **External** is selected, the instrument time base is synthesized using the clock provided externally to the Ext. Clock IN SMA connector.

External Clock Frequency[Hz]: when the External is selected, you should specify the External Clock Frequency value from 10 MHz to 100 MHz.

Remote / Locale Mode



The Remote/Local page allows you to enable or disable the VXI11 server. By default the VXI11 server is enabled and it means that the instrument can accept the SCPI commands that arrive from a remote machine.

When the instrument will receive SCPI commands, the UI will change accordingly.

Please note that you cannot control the Model 765 instrument sending SCPI commands in the local machine.

If the VXI-11 server will be disabled, the SCPI commands communication will be disabled.

Remote Desktop Connection

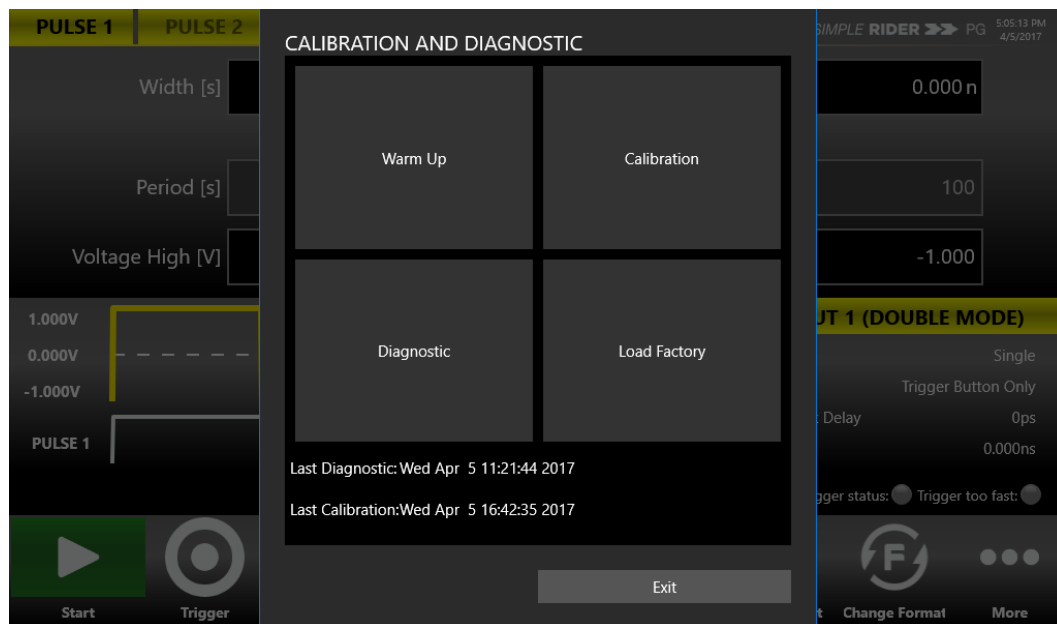
If you need to connect to the instrument using the remote desktop connection, you should insert the following credentials:

Computer Name: AT

User Name: AT

Password: 1234

Self Calibration and Diagnostic



The instrument performs a limited set of hardware tests at power-on.

You can also perform the Self Calibration and Self Diagnostic from the More →

Calibration/Diagnostic page pressing the button .

CAUTION. Do not power off the instrument while executing Self Calibration.

If the power is turned off during Self Calibration, data stored in the internal memory may be lost and the instrument may not work correctly: in this case you should reload the Factory Calibration parameters.

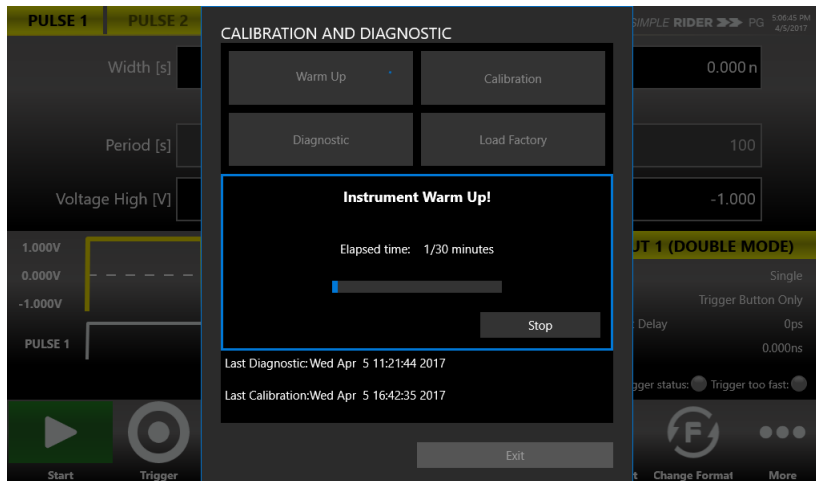
NOTE. Before executing the diagnostic or calibration operations, allow a 30 minute warm-up period after powering on the instrument, because the calibration is not valid if the instrument does not reach to a valid temperature.

Self Calibration

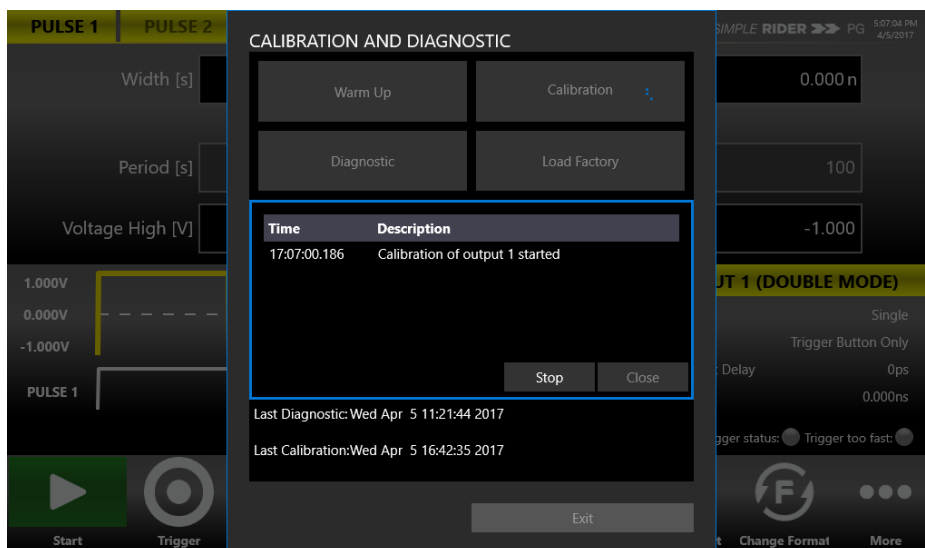
This calibration primarily checks DC accuracy using the internal calibration routines.



1. Click the More → Calibration and Diagnostic button .
2. Click the Warm up Timer, and one dialog will pop up to show warm up timer. Wait for 30 minutes. You can press Stop to terminate warming up.
3. When 30 minutes is shown on warm up timer dialog, press Stop.



4. Click the Self Calibration button to start the calibration and the following dialog will appear.

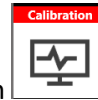


5. Self Calibration may take for more than 10 minutes. It can be stopped pressing the Stop button.
6. When the Self Calibration is complete, press the Close button.

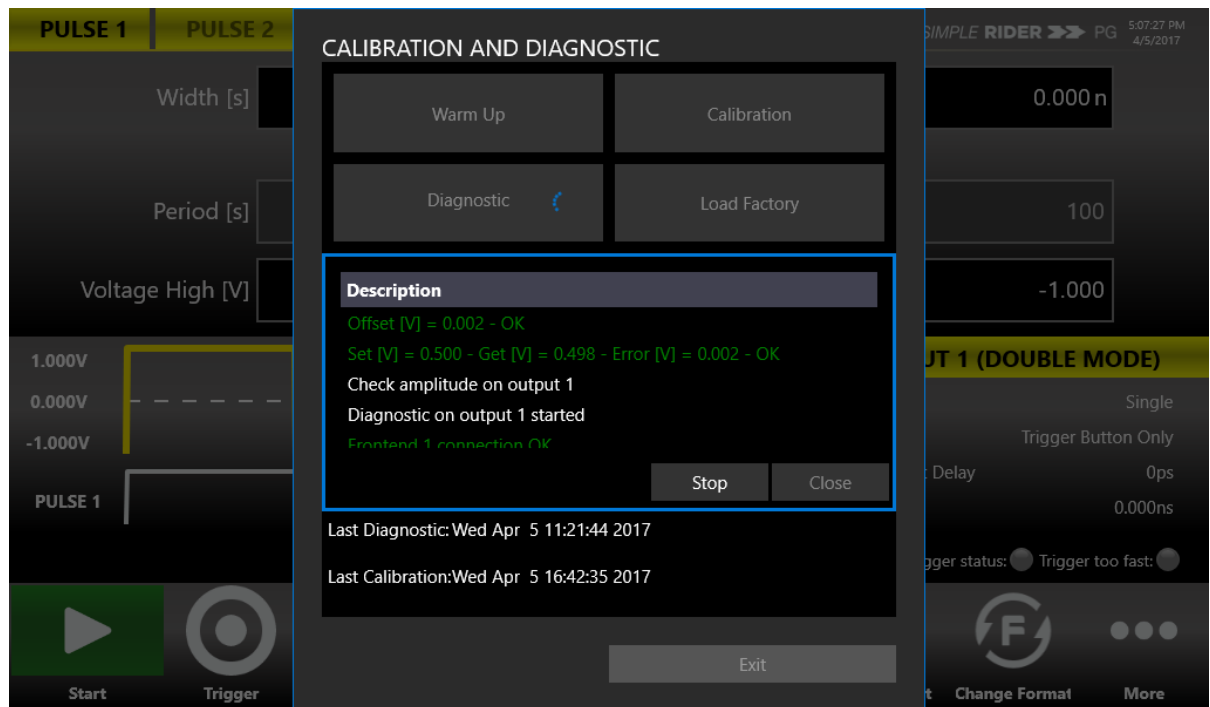
Self Diagnostic

This test verifies that your instrument is operating correctly.

NOTE. Before executing this operation, allow a 30 minute warm-up period after powering on the instrument, because the calibration is not valid if the instrument does not reach to a valid temperature.



1. Click the More → Calibration and Diagnostic button .
2. Click the Warm up Timer, and one dialog will pop up to show warm up timer. Wait for 30 minutes. You can press Stop to terminate warming up.
3. When 30 minutes is shown on warm up timer dialog, press Stop.
4. Click the Self Diagnostic button and the following dialog will appear and the diagnostic will start.



5. You can interrupt the self diagnostic pressing the Stop button.
6. During the diagnostic operations, the steps of this procedure are displayed under the Description section of this window.

Quick tips

- Allow a 30 minute warm-up period before executing Self Calibration or Self Diagnostic.
- Disconnect all the cables from the instrument when you perform Self Calibration or Self Diagnostic.
- It is recommended that the Self Calibration should be performed along with a periodic check.
- Don't power off the instrument during the Self Calibration or Self Diagnostic operations.
- If the power is turned off during Self Calibration, in this case you should reload the Factory Calibration parameters.



Contact Us

Berkeley Nucleonics Corporation
2955 Kerner Blvd.
San Rafael, CA 94901

Phone: (415) 453-9955
Email: info@berkeleynucleonics.com
Web: www.berkeleynucleonics.com

Model 765 User Manual

Document Version Number: Rev B, 2.3
Print Code: 02025060